A ZERO–HOPF ROUTE TO TORUS BIFURCATION IN THE HINDMARSH–ROSE SYSTEM

Abstract. We investigate a fixed-equilibrium unfolding of the three-dimensional polynomial Hindmarsh–Rose system near a zero–Hopf equilibrium and track the small periodic orbit born from that point to a secondary torus bifurcation. First-order averaging yields the periodic branch, but it does not locate the finite-ϵ Neimark–Sacker point because the transverse multipliers coalesce at 1 as $\epsilon \to 0$. To resolve this degeneracy, we derive the Poincaré map to second order, construct the local parameter curve on which the two nontrivial Floquet multipliers have exactly unit modulus, and prove a transverse crossing. Along this curve, the first Neimark–Sacker Lyapunov coefficient satisfies

$$l_1^\epsilon = \epsilon^2 L_{\mathrm{HR}} + \mathcal{O}(\epsilon^3),$$

with an explicit expression for L_{HR} in the model parameters. Its sign determines both the criticality of the bifurcation and the normal stability of the emerging invariant two-torus. Full-system computations in an explicit algebraic parameter regime confirm the asymptotic displacement of the critical curve, exhibit the attracting invariant closed curve on the canonical Poincaré section, and recover the expected square-root amplitude scaling. The analysis establishes a rigorous local zero–Hopf-to-torus mechanism for the specified generalized polynomial family; it does not claim that the conventional slow-adaptation neuronal parameter regime realizes the same mechanism.

Introduction

The Hindmarsh–Rose (HR) system is a three-dimensional polynomial model introduced to describe neuronal spiking and bursting [11]. Its relatively simple form permits both numerical investigation and analytical study of the bifurcations underlying oscillatory behavior. Studies of the classical model have characterized transitions between bursting and spiking chaos [12], analyzed its bifurcation structure [25], and identified codimension-two homoclinic bifurcations organizing spike adding [17]. More recently, Barrio, Ibáñez, and Pérez examined geometric changes in parameter slices of bifurcation sets in the HR and FitzHugh–Nagumo systems [3]. These results illustrate the importance of distinguishing the mechanisms responsible for different oscillatory transitions in neuronal models.

One such mechanism is a secondary Hopf, or torus, bifurcation of a periodic orbit. In a three-dimensional autonomous system, it corresponds to a Neimark–Sacker bifurcation of the associated two-dimensional Poincaré map: a nonreal pair of transverse Floquet multipliers crosses the unit circle, and an invariant closed curve gives rise to an invariant two-torus of the flow. Torus bifurcations and torus-canard dynamics have been studied in several neuronal bursters [4, 14], where they can organize transitions between tonic spiking and bursting. An invariant torus may support quasiperiodic motion, but its existence alone does not exclude resonant dynamics. Here, we study the local creation and normal stability of a torus from a small periodic orbit near a zero–Hopf equilibrium; global continuation, resonance tongues, and torus breakdown are outside the scope of the analysis.

A zero–Hopf equilibrium has one zero eigenvalue and a simple pair of purely imaginary eigenvalues. For the HR system, Buzzi, Llibre, and Medrado established Hopf and zero–Hopf bifurcations and used averaging to obtain nearby periodic solutions [5]. The

2020 *Mathematics Subject Classification.* Primary 37G10; Secondary 34C23, 34C25.

Key words and phrases. Hindmarsh–Rose system, zero–Hopf bifurcation, averaging, Neimark–Sacker bifurcation, Floquet multipliers, invariant two-tori.

present work addresses a subsequent question for a specified unfolding: under what explicit parameter conditions does the resulting periodic branch undergo a nondegenerate torus bifurcation, and what determines the side and stability of the bifurcating torus?

The general averaging framework for this question is already available. Cândido and Novaes developed conditions for torus bifurcations in two-parameter families of periodic differential equations by studying Neimark–Sacker bifurcations of their Poincaré maps [6]. Cândido, Novaes, and Valls subsequently applied higher-order averaging to periodic solutions and invariant tori in the Rössler system [8]. A particularly close model-specific precedent is the work of Cândido, Novaes, and Sadri on the FitzHugh–Nagumo system [7]. Their Family A treats a periodic orbit obtained by first-order averaging and a secondary torus bifurcation whose Lyapunov coefficient requires a second-order calculation after the leading term vanishes. They also investigate more degenerate parameter families using higher-order averaging and Lyapunov–Schmidt reduction. Normal hyperbolicity of tori arising in secondary Hopf bifurcations is addressed by Novaes and Pereira [19]. Our contribution concerns the explicit realization and classification of this established bifurcation mechanism in the chosen HR family, including its parameter conditions and quantitative asymptotics.

The higher-order calculation is needed because the leading averaged trace condition does not determine the finite-ϵ bifurcation parameter or the first nonzero criticality coefficient. At the leading critical parameter, the transverse multipliers satisfy

$$\mu_{\pm}(\epsilon) = 1 \pm 2\pi i \epsilon \nu_0 + \mathcal{O}(\epsilon^2), \qquad \nu_0 > 0.$$

The omitted second-order terms contribute at the same order as the modulus correction of the displayed approximation, so this expansion alone does not establish an exact unit-modulus condition. Nevertheless, a simple averaged zero and a transverse averaged eigenvalue crossing give a nearby exact critical curve through the implicit-function theorem. We compute the second-order return map to determine its first correction and to evaluate the first nonzero Neimark–Sacker Lyapunov term. Along the critical curve, the rotation angle tends to zero with ϵ, and the homological resolvents are of order $\mathcal{O}(\epsilon^{-1})$. Their dependence on ϵ must therefore be retained when extracting the coefficient.

We consider an exact parameter constraint that keeps the first coordinate of the distinguished equilibrium fixed while making the equilibrium explicit throughout the unfolding. Under the stated nondegeneracy conditions, the analysis gives the following results:

(i) An explicit simple zero of the first averaged function determines the small periodic branch, its leading amplitude and period, and its transverse stability away from the leading critical parameter.

(ii) On this local periodic branch, a unique exact unit-modulus curve is constructed and its first correction is computed:

$$\sigma_c(\epsilon) = \sigma_1^* + \epsilon\sigma_{1,1} + \mathcal{O}(\epsilon^2).$$

The transverse Floquet pair is nonreal and crosses the unit circle transversely on this curve.

(iii) The first Neimark–Sacker Lyapunov coefficient has the expansion

$$l_1^{\epsilon} = \epsilon^2 L_{\mathrm{HR}} + \mathcal{O}(\epsilon^3),$$

where L_{HR} is given by an explicit factorized expression in the HR parameters. The calculation verifies the cancellation of the order-ϵ term and shows that the leading nonzero coefficient is independent of the second-order unfolding coefficients β_2 and σ_2.

(iv) When $L_{\mathrm{HR}} \neq 0$, the local Neimark–Sacker theorem yields a branch of invariant two-tori and determines its existence side, criticality, and normal attraction or

repulsion. These conclusions hold for each fixed sufficiently small $\epsilon > 0$, in phase-space and parameter neighborhoods that may depend on ϵ.

Full-system computations in an explicit algebraic parameter regime provide quantitative checks of the local predictions. Shooting and variational integration locate the Floquet crossing, and a decreasing sequence of ϵ-values tests the predicted first correction to the critical curve. On the attracting-torus side, the canonical Poincaré section displays a closed curve surrounding the unstable periodic-orbit fixed point. The section amplitude is consistent with square-root growth near onset for fixed ϵ. These calculations support the predicted location, geometry, and criticality of the bifurcation; the existence argument is analytical.

The parameter regime requires a qualification. With $a, d, s_0 > 0$, the zero–Hopf relations used here force $r > 1$, $x_0 < 0$, and $b_0 < 0$, as explained in Remark 1.1. Thus the organizing equilibrium is outside the conventional HR regime with positive a, b, d, s, including the usual slow-adaptation range $0 < r \ll 1$. The results concern a generalized polynomial HR family and do not establish this mechanism for the conventional physiological parameter set. Changing only the unfolding does not remove this obstruction; a connection to that regime would require a modification or extension of the model.

The paper is organized as follows. Section 1 introduces the model, the zero–Hopf family, and the main results. Section 2 constructs the small periodic orbit by first-order averaging. Section 3 derives the critical curve, computes the Lyapunov coefficient, and proves the local torus bifurcation. Section 4 presents the full-system numerical checks, and Section 5 discusses the conclusions and limitations. Appendix A contains the higher-order coefficient calculation.

1. The Hindmarsh–Rose model and main results

1.1. **The Hindmarsh–Rose model.** We consider the three-dimensional Hindmarsh–Rose system [11]

$$\begin{aligned} \dot{x} &= y - ax^3 + bx^2 - z + I, \\ \dot{y} &= c - dx^2 - y, \\ \dot{z} &= r\big(s(x - x_R) - z\big), \end{aligned} \tag{1}$$

where $x, y, z \in \mathbb{R}$ and

$$a, b, c, d, r, s, x_R, I \in \mathbb{R}, \qquad r > 0.$$

In the neuronal interpretation, x represents the membrane potential, y is a fast recovery variable, and z is an adaptation variable. The parameter I represents the applied current, x_R is a reference potential, s determines the coupling between the membrane potential and the adaptation variable, and r controls the time scale of the z-dynamics. The usual slow-adaptation regime corresponds to $0 < r \ll 1$.

System (1) defines a smooth autonomous vector field on $\mathbb{R}^3$. An equilibrium $p_e = (x_e, y_e, z_e)$ satisfies

$$y_e = c - dx_e^2, \qquad z_e = s(x_e - x_R), \tag{2}$$

where x_e is a real root of

$$ax_e^3 + (d - b)x_e^2 + sx_e - (c + sx_R + I) = 0. \tag{3}$$

Thus every real root of (3) determines an equilibrium through (2). The Jacobian matrix of (1) is

$$DF(x, y, z) = \begin{pmatrix} -3ax^2 + 2bx & 1 & -1 \\ -2dx & -1 & 0 \\ rs & 0 & -r \end{pmatrix}. \tag{4}$$

1.2. **A zero–Hopf parameter family.** Buzzi, Llibre, and Medrado previously analyzed zero–Hopf bifurcations in the Hindmarsh–Rose system [5]. Here we introduce a parameter family for (1) that is tailored to the higher-order analysis required below.

Fix $x_0 \neq 0$, while a, c, d, r, and x_R are held fixed, and define

$$\begin{aligned} b_0 &= \frac{1+r+3ax_0^2}{2x_0}, & s_0 &= 1+r-2dx_0, \\ I_0 &= -c - dx_0^2 - \frac{a}{2}x_0^3 + \frac{1+r}{2}x_0 - s_0 x_R. \end{aligned} \tag{5}$$

For $(b, s, I) = (b_0, s_0, I_0)$, equations (2)–(3) give the equilibrium

$$p_0 = \big(x_0, c - dx_0^2, s_0(x_0 - x_R)\big). \tag{6}$$

The linearization of (1) at p_0 is

$$A_0 = \begin{pmatrix} 1+r & 1 & -1 \\ -2dx_0 & -1 & 0 \\ rs_0 & 0 & -r \end{pmatrix}. \tag{7}$$

Its characteristic polynomial reduces to

$$\det(\lambda I - A_0) = \lambda(\lambda^2 + \omega^2), \qquad \omega^2 = 2dx_0(1-r) - 1. \tag{8}$$

Throughout the paper, we assume that

$$2dx_0(1-r) > 1 \tag{9}$$

and set

$$\omega = \sqrt{2dx_0(1-r) - 1} > 0. \tag{10}$$

It follows that

$$\operatorname{spec}(A_0) = \{0, \pm i\omega\},$$

and hence p_0 is a zero–Hopf equilibrium. In particular, (9) excludes $r = 1$.

For later use, we record the equivalent relations

$$2dx_0 = \frac{1+\omega^2}{1-r}, \qquad s_0 = \frac{r^2+\omega^2}{r-1}. \tag{11}$$

Remark 1.1 (Parameter-regime restriction). Under the conventional sign assumptions $d > 0$ and $s_0 > 0$, (11) imply $r > 1$ and $x_0 < 0$. If, in addition, $a > 0$, then (5) gives $b_0 < 0$. Consequently, this zero–Hopf family does not lie in the commonly used slow-adaptation regime with $a, b, d, s > 0$ and $0 < r \ll 1$. The results below are therefore a rigorous analysis of the generalized polynomial Hindmarsh–Rose family, rather than a direct statement about the canonical slow-adaptation neuronal parameter set.

1.3. **The fixed-equilibrium unfolding.** To unfold the zero–Hopf equilibrium while keeping its first coordinate fixed, we consider

$$b(\epsilon) = b_0 + \epsilon\beta_1 + \epsilon^2\beta_2, \qquad s(\epsilon) = s_0 + \epsilon\sigma_1 + \epsilon^2\sigma_2, \tag{12}$$

and impose the exact constraint

$$I(\epsilon) = -c + dx_0^2 + ax_0^3 - b(\epsilon)x_0^2 + s(\epsilon)(x_0 - x_R). \tag{13}$$

Then

$$p_\epsilon = \big(x_0, c - dx_0^2, s(\epsilon)(x_0 - x_R)\big) \tag{14}$$

is an equilibrium of the unfolded system for all sufficiently small ϵ, and $p_\epsilon \to p_0$ as $\epsilon \to 0$.

We introduce the quantities

$$k_0 := b_0 - 3ax_0 = \frac{1+r-3ax_0^2}{2x_0}, \qquad D_0 := k_0(\omega^2 - r) + rd. \tag{15}$$

The first-order averaged coefficients used below are

$$\begin{aligned} A_X &= 2x_0\beta_1\left(1-\frac{r}{\omega^2}\right)+\frac{r\sigma_1}{\omega^2}, & A_Z &= \frac{r(2x_0\beta_1-\sigma_1)}{\omega^2},\\ B_X &= k_0\left(1-\frac{r}{\omega^2}\right)+\frac{rd}{\omega^2}=\frac{D_0}{\omega^2}, & B_Y &= \frac{d-(1+r)k_0}{\omega},\\ B_Z &= \frac{r(k_0-d)}{\omega^2}, & E_X &= -a\left(1-\frac{r}{\omega^2}\right)=-\frac{a(\omega^2-r)}{\omega^2}. \end{aligned} \tag{16}$$

When $k_0 \neq 0$, define the leading-order critical parameter

$$\sigma_1^* := \frac{2dx_0}{k_0}\,\beta_1. \tag{17}$$

Finally, let

$$P_{\mathrm{HR}} := 3a\omega^2(\omega^2-r)+\big(2k_0\omega^2+9r(d-k_0)\big)\big(d-k_0(1+r)\big), \tag{18}$$

and

$$L_{\mathrm{HR}} := -\frac{\pi k_0 r(k_0-d)}{2\omega^3 D_0^2}\,P_{\mathrm{HR}}. \tag{19}$$

1.4. **Main results.** The first theorem establishes the existence of a small-amplitude periodic orbit bifurcating from the zero–Hopf equilibrium.

Theorem 1.2 (Periodic orbit near the zero–Hopf equilibrium)**.** *Consider the Hindmarsh–Rose system* (1) *under the exact fixed-equilibrium unfolding* (12)–(13)*. Assume* (9) *and*

$$B_X B_Z \neq 0, \qquad \frac{A_X(2B_XA_Z-A_XB_Z)}{B_Z} > 0. \tag{20}$$

Then, for every sufficiently small $\epsilon \neq 0$, the unfolded system possesses a periodic orbit Γ_ϵ satisfying

$$\operatorname{dist}(\Gamma_\epsilon, p_0) = \mathcal{O}(|\epsilon|). \tag{21}$$

Its period satisfies

$$T_\epsilon = \frac{2\pi}{\omega}+\mathcal{O}(\epsilon). \tag{22}$$

Moreover, after a suitable choice of phase, the leading-order parametrization of Γ_ϵ is obtained explicitly in Section 2.4.

Section 2 proves Theorem 1.2 by combining a real canonical reduction of the zero–Hopf linearization with a zero–Hopf blow-up and first-order averaging. The transverse stability of the resulting periodic branch is analyzed in Section 3.1.

The second theorem follows this local periodic-orbit family to a nondegenerate Neimark–Sacker bifurcation and characterizes the resulting invariant two-tori. The quantities ν_0 and K_* appearing in the first correction of the critical parameter are defined explicitly in Section 3.4.

Theorem 1.3 (Neimark–Sacker bifurcation of invariant two-tori)**.** *Consider the exact fixed-equilibrium unfolding* (12)–(13)*. Assume that*

$$2dx_0(1-r)>1, \qquad k_0D_0\neq 0, \qquad B_XB_Z<0, \qquad A_X(\sigma_1^*)\neq 0, \qquad P_{\mathrm{HR}}\neq 0. \tag{23}$$

Then there exists $\epsilon_0>0$ such that, for every $0<\epsilon<\epsilon_0$, the following conclusions hold.

(1) *The local periodic-orbit family contains a unique critical periodic orbit $\Gamma_{\sigma_c(\epsilon),\epsilon}$, where*

$$\sigma_c(\epsilon) = \sigma_1^*+\epsilon\,\sigma_{1,1}+\mathcal{O}(\epsilon^2), \qquad \sigma_1^* = \frac{2dx_0}{k_0}\beta_1, \tag{24}$$

with

$$\sigma_{1,1} = \frac{D_0\omega}{rk_0}\left(\operatorname{tr} K_* + 2\pi\nu_0^2\right). \tag{25}$$

At $\sigma_1 = \sigma_c(\epsilon)$, the two transverse Floquet multipliers form a nonreal conjugate pair of unit modulus. Moreover, the crossing is transverse, with

$$\left.\frac{\partial|\mu_\pm(\sigma_1,\epsilon)|}{\partial\sigma_1}\right|_{\sigma_1=\sigma_c(\epsilon)} = -\frac{\pi rk_0}{D_0\omega}\,\epsilon + \mathcal{O}(\epsilon^2) \neq 0. \tag{26}$$

(2) *The critical periodic orbit $\Gamma_{\sigma_c(\epsilon),\epsilon}$ undergoes a nondegenerate Neimark–Sacker bifurcation. A unique local branch of invariant two-tori bifurcates from this periodic orbit and exists precisely on the parameter side*

$$P_{\mathrm{HR}}\big(\sigma_1 - \sigma_c(\epsilon)\big) > 0. \tag{27}$$

In a sufficiently small neighborhood of the critical periodic orbit, no member of this local bifurcating branch exists on the opposite side of the critical parameter.

(3) *Along the critical curve, the first Neimark–Sacker Lyapunov coefficient satisfies*

$$l_1^\epsilon = \epsilon^2 L_{\mathrm{HR}} + \mathcal{O}(\epsilon^3). \tag{28}$$

Consequently, if $L_{\mathrm{HR}} < 0$, the bifurcation is supercritical and the bifurcating invariant two-torus is normally attracting. If $L_{\mathrm{HR}} > 0$, the bifurcation is subcritical and the bifurcating invariant two-torus is normally repelling. On the torus-existence side, the nearby periodic orbit has the opposite transverse stability.

Under the hypotheses of Theorem 1.3, the prefactor relating L_{HR} and P_{HR} is nonzero; hence

$$L_{\mathrm{HR}} \neq 0 \quad\Longleftrightarrow\quad P_{\mathrm{HR}} \neq 0.$$

Thus the condition $P_{\mathrm{HR}} \neq 0$ in (23) is precisely the nondegeneracy condition for the first nonvanishing Neimark–Sacker Lyapunov coefficient.

Section 3 proves Theorem 1.3. The proof first constructs the exact finite-ϵ unit-modulus curve and its first correction from the second-order Poincaré map, then verifies the transverse crossing and the low-order nonresonance conditions, and finally evaluates the first nonvanishing Neimark–Sacker Lyapunov coefficient. Suspending the bifurcating invariant closed curves of the Poincaré map yields the invariant two-tori of the continuous-time flow.

2. Periodic orbits near the zero–Hopf equilibrium

Fix a, c, d, r, x_R, and x_0, and use the parameter expansions (12) and (13). By construction, (14) is an equilibrium for all sufficiently small ϵ, with $p_\epsilon \to p_0$ as $\epsilon \to 0$. In particular,

$$\begin{aligned} I(\epsilon) = I_0 &+ \epsilon\big[-x_0^2\beta_1 + (x_0 - x_R)\sigma_1\big] \\ &+ \epsilon^2\big[-x_0^2\beta_2 + (x_0 - x_R)\sigma_2\big]. \end{aligned} \tag{29}$$

Translate the equilibrium p_ϵ to the origin by setting

$$x = x_0 + u, \qquad y = c - dx_0^2 + v, \qquad z = s(\epsilon)(x_0 - x_R) + w.$$

Then the system becomes exactly

$$\begin{aligned} \dot u &= (1+r)u + v - w + k_0u^2 - au^3 \\ &\quad + \epsilon\beta_1(2x_0u + u^2) + \epsilon^2\beta_2(2x_0u + u^2), \\ \dot v &= -2dx_0u - v - du^2, \\ \dot w &= rs_0u - rw + \epsilon r\sigma_1u + \epsilon^2 r\sigma_2u. \end{aligned} \tag{30}$$

The linear part of (30) at $\epsilon = 0$ is precisely the zero–Hopf matrix A_0, whereas the terms proportional to β_j and σ_j describe the chosen unfolding direction.

2.1. **Real canonical form of the linear part.** Let A_0 denote the linearization at the zero–Hopf equilibrium. Using (11), we choose a real basis consisting of the real and imaginary parts of an eigenvector for $i\omega$, together with an eigenvector for the zero eigenvalue. The corresponding transformation matrix is

$$P = \begin{pmatrix} 1 & 0 & 1 \\ -\dfrac{2dx_0}{1+\omega^2} & -\dfrac{2dx_0\omega}{1+\omega^2} & -2dx_0 \\ \dfrac{r^2 s_0}{r^2+\omega^2} & \dfrac{r s_0 \omega}{r^2+\omega^2} & s_0 \end{pmatrix}. \tag{31}$$

Since

$$\det P = \frac{\omega^3}{1-r} \neq 0,$$

the change of variables

$$\begin{pmatrix} u \\ v \\ w \end{pmatrix} = P \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} \tag{32}$$

is nonsingular and reduces the linear part to

$$P^{-1} A_0 P = \begin{pmatrix} 0 & -\omega & 0 \\ \omega & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{33}$$

The inverse transformation is

$$P^{-1} = \begin{pmatrix} 1 - \dfrac{r}{\omega^2} & -\dfrac{r}{\omega^2} & \dfrac{1}{\omega^2} \\ -\dfrac{1+r}{\omega} & -\dfrac{1}{\omega} & \dfrac{1}{\omega} \\ \dfrac{r}{\omega^2} & \dfrac{r}{\omega^2} & -\dfrac{1}{\omega^2} \end{pmatrix}. \tag{34}$$

We denote the rows of P^{-1} by ℓ_X, ℓ_Y, ℓ_Z, respectively. Equation (32) then gives

$$u = X + Z.$$

2.2. **Zero–Hopf blow-up.** To resolve solutions that bifurcate from the zero–Hopf equilibrium, we introduce the rescaling

$$(X, Y, Z) = \epsilon(x_1, x_2, x_3), \qquad U := x_1 + x_3,$$

so that $u = \epsilon U$. Substitution into (30), followed by (32), gives

$$\begin{aligned} \dot{x}_1 &= -\omega x_2 + \epsilon H_{11} + \epsilon^2 H_{21}, \\ \dot{x}_2 &= \omega x_1 + \epsilon H_{12} + \epsilon^2 H_{22}, \\ \dot{x}_3 &= \epsilon H_{13} + \epsilon^2 H_{23}, \end{aligned} \tag{35}$$

where

$$\begin{aligned} \begin{pmatrix} H_{11} \\ H_{12} \\ H_{13} \end{pmatrix} &= P^{-1}(m_1 U + n_2 U^2), \\ \begin{pmatrix} H_{21} \\ H_{22} \\ H_{23} \end{pmatrix} &= P^{-1}(m_2 U + n_{2,1} U^2 + n_3 U^3), \end{aligned} \tag{36}$$

with

$$m_1 = (2x_0\beta_1, 0, r\sigma_1)^{\mathsf{T}}, \quad n_2 = (k_0, -d, 0)^{\mathsf{T}},$$
$$m_2 = (2x_0\beta_2, 0, r\sigma_2)^{\mathsf{T}}, \quad n_{2,1} = (\beta_1, 0, 0)^{\mathsf{T}}, \quad n_3 = (-a, 0, 0)^{\mathsf{T}}.$$

For $j \in \{X, Y, Z\}$, define

$$A_j = \ell_j m_1, \qquad B_j = \ell_j n_2, \qquad C_j = \ell_j m_2, \qquad D_j = \ell_j n_{2,1}, \qquad E_j = \ell_j n_3.$$

Then

$$\begin{aligned} H_{11} &= A_X U + B_X U^2, & H_{21} &= C_X U + D_X U^2 + E_X U^3, \\ H_{12} &= A_Y U + B_Y U^2, & H_{22} &= C_Y U + D_Y U^2 + E_Y U^3, \\ H_{13} &= A_Z U + B_Z U^2, & H_{23} &= C_Z U + D_Z U^2 + E_Z U^3. \end{aligned} \tag{37}$$

The first-order coefficients are

$$\begin{aligned} A_X &= 2x_0\beta_1\left(1 - \frac{r}{\omega^2}\right) + \frac{r\sigma_1}{\omega^2}, & B_X &= k_0\left(1 - \frac{r}{\omega^2}\right) + \frac{rd}{\omega^2}, \\ A_Y &= \frac{r\sigma_1 - 2x_0(1+r)\beta_1}{\omega}, & B_Y &= \frac{d - (1+r)k_0}{\omega}, \\ A_Z &= \frac{r(2x_0\beta_1 - \sigma_1)}{\omega^2}, & B_Z &= \frac{r(k_0 - d)}{\omega^2}. \end{aligned} \tag{38}$$

Similarly, the second-order coefficients are

$$\begin{aligned} C_X &= 2x_0\beta_2\left(1 - \frac{r}{\omega^2}\right) + \frac{r\sigma_2}{\omega^2}, & D_X &= \beta_1\left(1 - \frac{r}{\omega^2}\right), & E_X &= -a\left(1 - \frac{r}{\omega^2}\right), \\ C_Y &= \frac{r\sigma_2 - 2x_0(1+r)\beta_2}{\omega}, & D_Y &= -\frac{(1+r)\beta_1}{\omega}, & E_Y &= \frac{(1+r)a}{\omega}, \\ C_Z &= \frac{r(2x_0\beta_2 - \sigma_2)}{\omega^2}, & D_Z &= \frac{r\beta_1}{\omega^2}, & E_Z &= -\frac{ar}{\omega^2}. \end{aligned} \tag{39}$$

2.3. **Cylindrical coordinates and first-order averaging.** Introduce cylindrical coordinates by

$$x_1 = \rho\cos\theta, \qquad x_2 = \rho\sin\theta, \qquad x_3 = \zeta, \qquad \rho > 0,$$

so that $U = \rho\cos\theta + \zeta$. For $k = 1, 2$, set

$$R_k = \cos\theta\, H_{k1} + \sin\theta\, H_{k2}, \qquad Q_k = \frac{-\sin\theta\, H_{k1} + \cos\theta\, H_{k2}}{\rho}, \qquad Z_k = H_{k3}.$$

Then (35) takes the form

$$\begin{aligned} \dot\rho &= \epsilon R_1 + \epsilon^2 R_2, \\ \dot\theta &= \omega + \epsilon Q_1 + \epsilon^2 Q_2, \\ \dot\zeta &= \epsilon Z_1 + \epsilon^2 Z_2. \end{aligned} \tag{40}$$

On every compact subset of $(0, \infty)\times\mathbb{R}$, the angular velocity remains nonzero for sufficiently small ϵ. Hence θ can be used as the independent variable. For $q = (\rho, \zeta)^{\mathsf{T}}$, the first-order reduced equation is

$$\frac{dq}{d\theta} = \epsilon F_1(\theta, q) + \mathcal{O}(\epsilon^2), \qquad F_1(\theta, q) = \frac{1}{\omega}\begin{pmatrix} R_1 \\ Z_1 \end{pmatrix}. \tag{41}$$

The normalized first averaged function is

$$g_1(q) = \frac{1}{2\pi}\int_0^{2\pi} F_1(\theta, q)\, d\theta.$$

Direct integration gives

$$g_1(\rho,\zeta)=\frac{1}{\omega}\begin{pmatrix}\rho\left(\dfrac{A_X}{2}+B_X\zeta\right)\\ A_Z\zeta+B_Z\left(\dfrac{\rho^2}{2}+\zeta^2\right)\end{pmatrix}. \tag{42}$$

2.4. **Proof of Theorem 1.2.** A zero (ρ_*,ζ_*) of the first averaged function (42) with $\rho_*>0$ corresponds to a nontrivial periodic solution of the averaged system. If $B_XB_Z\neq 0$, the equations $g_1(\rho,\zeta)=0$ admit the solution

$$\zeta_*=-\frac{A_X}{2B_X},\qquad \rho_*=\left[\frac{A_X(2B_XA_Z-A_XB_Z)}{2B_X^2B_Z}\right]^{1/2}. \tag{43}$$

This solution is admissible precisely under (20). At (ρ_*,ζ_*), the Jacobian matrix of g_1 is

$$Dg_1(\rho_*,\zeta_*)=\frac{1}{\omega}\begin{pmatrix}0 & B_X\rho_*\\ B_Z\rho_* & A_Z-\dfrac{A_XB_Z}{B_X}\end{pmatrix}. \tag{44}$$

Consequently,

$$\det Dg_1(\rho_*,\zeta_*)=-\frac{B_XB_Z\rho_*^2}{\omega^2}\neq 0. \tag{45}$$

Thus (ρ_*,ζ_*) is a simple zero of the first averaged function.

The reduced system is 2π-periodic in θ. Under (20), (ρ_*,ζ_*) is an admissible simple zero. The first-order averaging theorem [23, Chapters 2–3] therefore gives, for every sufficiently small $\epsilon\neq 0$, a 2π-periodic solution of the reduced system near (ρ_*,ζ_*). Since

$$\dot\theta=\omega+\mathcal{O}(\epsilon),$$

this solution determines a periodic orbit Γ_ϵ of the original Hindmarsh–Rose system, whose period satisfies

$$T_\epsilon=\frac{2\pi}{\omega}+\mathcal{O}(\epsilon).$$

The zero–Hopf blow-up and the inverse coordinate transformations give

$$\Gamma_\epsilon=p_\epsilon+\mathcal{O}(\epsilon).$$

Since $p_\epsilon\to p_0$ as $\epsilon\to 0$, it follows that $\operatorname{dist}(\Gamma_\epsilon,p_0)=\mathcal{O}(|\epsilon|)$, which proves Theorem 1.2.

Let

$$\Omega_\epsilon:=\frac{2\pi}{T_\epsilon}=\omega+\mathcal{O}(\epsilon).$$

After a suitable phase choice, the periodic orbit has the following leading-order parametrization:

$$\begin{aligned}
x(t,\epsilon)&=x_0+\epsilon\big[\rho_*\cos(\Omega_\epsilon t)+\zeta_*\big]+\mathcal{O}(\epsilon^2),\\
y(t,\epsilon)&=c-dx_0^2\\
&\quad-2dx_0\epsilon\left[\frac{\rho_*\cos(\Omega_\epsilon t)+\omega\rho_*\sin(\Omega_\epsilon t)}{1+\omega^2}+\zeta_*\right]+\mathcal{O}(\epsilon^2),\\
z(t,\epsilon)&=s(\epsilon)(x_0-x_R)\\
&\quad+s_0\epsilon\left[\frac{r^2\rho_*\cos(\Omega_\epsilon t)+r\omega\rho_*\sin(\Omega_\epsilon t)}{r^2+\omega^2}+\zeta_*\right]+\mathcal{O}(\epsilon^2).
\end{aligned} \tag{46}$$

Equivalently, the parametrization may be written in terms of an angular phase $\theta\in[0,2\pi]$. This formulation avoids the secular phase error that would result from replacing the actual frequency Ω_ϵ by ω in a representation intended to remain uniform in time.

3. Neimark–Sacker bifurcation and invariant two-tori

This section proves Theorem 1.3. We first examine the transverse stability of the periodic orbit from Theorem 1.2. We then construct the exact unit-modulus parameter curve, compute the first Neimark–Sacker Lyapunov coefficient, and apply the local Neimark–Sacker theorem. Suspending the resulting invariant closed curves produces invariant two-tori of the flow.

3.1. **First-order transverse stability.** Let Π_ϵ denote the local θ-return map associated with the periodic orbit furnished by Theorem 1.2. Locally in C^1,

$$\Pi_\epsilon(q) = q + 2\pi\epsilon g_1(q) + \mathcal{O}_{C^1}(\epsilon^2). \tag{47}$$

Hence, if λ_1, λ_2 are the eigenvalues of $Dg_1(\rho_*, \zeta_*)$, then the two eigenvalues of the derivative of the return map satisfy

$$\mu_j(\epsilon) = 1 + 2\pi\epsilon\lambda_j + \mathcal{O}(\epsilon^2), \qquad j = 1, 2. \tag{48}$$

These are precisely the two nontrivial, or transverse, Floquet multipliers of the periodic orbit. The full three-dimensional variational flow has, in addition, the trivial Floquet multiplier 1 associated with the flow direction.

Set

$$\tau := \operatorname{tr} Dg_1(\rho_*, \zeta_*) = \frac{B_X A_Z - A_X B_Z}{B_X \omega}, \qquad \Delta := \det Dg_1(\rho_*, \zeta_*) = -\frac{B_X B_Z \rho_*^2}{\omega^2}. \tag{49}$$

Suppose first that $B_X B_Z > 0$. Then $\Delta < 0$, so the eigenvalues λ_1, λ_2 are real and have opposite signs. It follows from (48) that, for every sufficiently small $\epsilon \neq 0$, one transverse Floquet multiplier lies inside the unit circle while the other lies outside it. Hence Γ_ϵ is of saddle type.

Suppose now that $B_X B_Z < 0$. Then $\Delta > 0$. If $\tau \neq 0$, the two eigenvalues of $Dg_1(\rho_*, \zeta_*)$ have real parts with the sign of τ. Indeed, if the eigenvalues are real, their positive product implies that they have the same sign, which is determined by their sum τ; if they form a nonreal conjugate pair, each has real part $\tau/2$. Therefore,

$$\epsilon\tau < 0 \quad \Longrightarrow \quad |\mu_1(\epsilon)| < 1, \qquad |\mu_2(\epsilon)| < 1$$

for sufficiently small $|\epsilon|$, and the periodic orbit is orbitally asymptotically stable. Conversely,

$$\epsilon\tau > 0 \quad \Longrightarrow \quad |\mu_1(\epsilon)| > 1, \qquad |\mu_2(\epsilon)| > 1,$$

so the periodic orbit is transversely repelling.

The remaining case

$$B_X B_Z < 0, \qquad \tau = 0,$$

is precisely the leading-order candidate for a Neimark–Sacker bifurcation. In this case, the eigenvalues of the first averaged Jacobian form a nonzero purely imaginary conjugate pair. Consequently, (48) does not determine whether the exact transverse multipliers lie inside, outside, or on the unit circle, because the first nontrivial correction to their modulus occurs at order ϵ^2. The second-order Poincaré-map calculation developed below is therefore necessary to locate the exact unit-modulus parameter curve and to determine the corresponding torus bifurcation.

3.2. **Leading-order Neimark–Sacker candidate.** We regard σ_1 as the bifurcation parameter, while all other unfolding parameters remain fixed. Recall that the trace and determinant of the Jacobian of the first averaged function at its nontrivial zero $q_* = (\rho_*, \zeta_*)^{\mathsf{T}}$ are

$$\tau(\sigma_1) = \operatorname{tr} Dg_1(q_*; \sigma_1), \qquad \Delta(\sigma_1) = \det Dg_1(q_*; \sigma_1) = -\frac{B_X B_Z \rho_*^2}{\omega^2}.$$

A leading-order candidate for a Neimark–Sacker bifurcation therefore requires

$$B_X B_Z < 0, \qquad \tau(\sigma_1) = 0,$$

so that $Dg_1(q_*;\sigma_1)$ has a nonzero purely imaginary conjugate pair. From the definitions of the averaged coefficients,

$$\tau(\sigma_1) = \frac{r(2dx_0\beta_1 - k_0\sigma_1)}{D_0\omega}. \tag{50}$$

Assume that $k_0 D_0 \neq 0$. Then the unique zero of τ is

$$\sigma_1^* = \frac{2dx_0}{k_0}\beta_1, \qquad \tau'(\sigma_1^*) = -\frac{rk_0}{D_0\omega} \neq 0. \tag{51}$$

Thus the conjugate eigenvalue pair crosses the imaginary axis transversely at the level of the first averaged system.

Suppose, in addition, that

$$B_X B_Z < 0, \qquad A_X(\sigma_1^*) \neq 0.$$

At $\sigma_1 = \sigma_1^*$, the nontrivial zero of g_1 is given by

$$\zeta_* = -\frac{A_X}{2B_X}, \qquad \rho_*^2 = \frac{A_X^2}{2B_X^2} = 2\zeta_*^2 > 0, \tag{52}$$

where the coefficients are evaluated at $\sigma_1 = \sigma_1^*$. Moreover,

$$\det Dg_1(q_*;\sigma_1^*) = -\frac{B_X B_Z \rho_*^2}{\omega^2} > 0,$$

so q_* is a simple zero. The eigenvalues of the averaged Jacobian are therefore

$$\lambda_\pm = \pm i\nu_0, \qquad \nu_0 := \frac{\rho_*\sqrt{-B_X B_Z}}{\omega} > 0. \tag{53}$$

The corresponding transverse multipliers of the return map satisfy

$$\mu_\pm(\epsilon) = 1 \pm 2\pi i\epsilon\nu_0 + \mathcal{O}(\epsilon^2). \tag{54}$$

This expansion identifies only the leading-order critical parameter. In particular, it does not imply $|\mu_\pm(\epsilon)| = 1$, since

$$|1 \pm 2\pi i\epsilon\nu_0|^2 = 1 + 4\pi^2\epsilon^2\nu_0^2.$$

Accordingly, the exact unit-modulus curve cannot be determined from first-order averaging alone and must be obtained from the second-order expansion of the Poincaré map.

3.3. **Second-order averaging.** Let $q = (\rho, \zeta)^\mathsf{T}$. On a compact neighborhood of the nontrivial averaged zero, contained in $(0,\infty)\times\mathbb{R}$, the angular velocity remains nonzero for sufficiently small ϵ. Using θ as the independent variable therefore gives

$$\frac{dq}{d\theta} = \epsilon F_1(\theta, q; \sigma_1) + \epsilon^2 F_2(\theta, q; \sigma_1) + \epsilon^3 R(\theta, q, \sigma_1, \epsilon), \tag{55}$$

where R is 2π-periodic in θ and smooth jointly in its remaining arguments. Here and below we use the exact quadratic fixed-equilibrium unfolding (12), with $I(\epsilon)$ determined by (13). The second-order term is

$$F_2 = \begin{pmatrix} \dfrac{R_2}{\omega} - \dfrac{R_1 Q_1}{\omega^2} \\[2ex] \dfrac{Z_2}{\omega} - \dfrac{Z_1 Q_1}{\omega^2} \end{pmatrix}. \tag{56}$$

In particular, the contributions involving Q_1, arising from the change of independent variable, are essential to the second-order calculation.

With the normalization adopted for g_1, define

$$g_2(q;\sigma_1) = \frac{1}{2\pi}\int_0^{2\pi}\left[F_2(\theta,q;\sigma_1) + D_qF_1(\theta,q;\sigma_1)\int_0^{\theta}F_1(\phi,q;\sigma_1)\,d\phi\right]d\theta. \tag{57}$$

Writing $g_2 = (g_{21}, g_{22})^{\mathsf{T}}$, direct integration yields

$$g_{21}(\rho,\zeta) = \frac{G_{21}(\rho,\zeta)}{24\omega^2}, \qquad g_{22}(\rho,\zeta) = \frac{G_{22}(\rho,\zeta)}{12\omega^2}, \tag{58}$$

where

$$\begin{aligned} G_{21} = {} & 6\pi A_X^2\rho - 6A_XA_Y\rho + 12A_XA_Y\zeta + 24\pi A_XB_X\rho\zeta - 4A_XB_Y\rho^2 \\ & - 12A_XB_Y\rho\zeta + 12A_XB_Y\zeta^2 + 12A_YA_Z\rho - 24A_YA_Z\zeta \\ & - 36A_YB_X\rho\zeta + 24A_YB_X\zeta^2 - 12A_YB_Z\rho^2 + 24A_YB_Z\rho\zeta - 24A_YB_Z\zeta^2 \\ & + 24\pi A_ZB_X\rho\zeta + 12A_ZB_Y\rho\zeta - 48A_ZB_Y\zeta^2 + 24\pi B_X^2\rho\zeta^2 \\ & - 6B_XB_Y\rho^3 - 8B_XB_Y\rho^2\zeta - 48B_XB_Y\rho\zeta^2 + 24B_XB_Y\zeta^3 \\ & + 12\pi B_XB_Z\rho^3 + 24\pi B_XB_Z\rho\zeta^2 - 3B_YB_Z\rho^3 - 24B_YB_Z\rho^2\zeta \\ & + 36B_YB_Z\rho\zeta^2 - 48B_YB_Z\zeta^3 + 12C_X\omega\rho + 24D_X\omega\rho\zeta \\ & + 9E_X\omega\rho^3 + 36E_X\omega\rho\zeta^2, \end{aligned} \tag{59}$$

and

$$\begin{aligned} G_{22} = {} & 6\pi A_XB_Z\rho^2 - 18A_YA_Z\zeta - 3A_YB_Z\rho^2 + 12A_YB_Z\rho\zeta - 30A_YB_Z\zeta^2 \\ & + 12\pi A_Z^2\zeta - 6A_ZB_Y\rho^2 - 24A_ZB_Y\zeta^2 + 6\pi A_ZB_Z\rho^2 + 36\pi A_ZB_Z\zeta^2 \\ & + 12\pi B_XB_Z\rho^2\zeta + 4B_YB_Z\rho^3 - 18B_YB_Z\rho^2\zeta + 12B_YB_Z\rho\zeta^2 \\ & - 36B_YB_Z\zeta^3 + 12\pi B_Z^2\rho^2\zeta + 24\pi B_Z^2\zeta^3 + 12C_Z\omega\zeta \\ & + 6D_Z\omega\rho^2 + 12D_Z\omega\zeta^2 + 18E_Z\omega\rho^2\zeta + 12E_Z\omega\zeta^3. \end{aligned} \tag{60}$$

Here the coefficients are those given in (38)–(39); their dependence on σ_1 is left implicit.

Lemma 3.1. *Let Π_{ϵ,σ_1} be the θ-return map of (55). There exist a bounded open neighborhood $U \Subset (0,\infty)\times\mathbb{R}$ of q_*, an interval J_0 containing σ_1^*, and $\epsilon_0 > 0$ such that*

$$\Pi_{\epsilon,\sigma_1}(q) = q + 2\pi\epsilon g_1(q;\sigma_1) + 2\pi\epsilon^2 g_2(q;\sigma_1) + \epsilon^3\mathcal{R}(q,\sigma_1,\epsilon) \tag{61}$$

for $(q,\sigma_1,\epsilon) \in U\times J_0\times[-\epsilon_0,\epsilon_0]$. The remainder $\mathcal{R}$ is C^4 jointly in (q,σ_1,ϵ), and it is uniformly bounded together with all derivatives of order at most four.

Proof. Let $q(\theta;q_0,\sigma_1,\epsilon)$ denote the solution of (55) satisfying $q(0;q_0,\sigma_1,\epsilon) = q_0$. The smooth-dependence theorem for ordinary differential equations implies that this solution is C^∞ jointly in $(\theta,q_0,\sigma_1,\epsilon)$ while it remains in the chosen compact neighborhood. By reducing U, J_0, ϵ_0, all solutions under consideration exist on $[0,2\pi]$, remain in the region where the angular velocity is nonzero, and depend smoothly on the initial condition and parameters.

At $\epsilon = 0$, the reduced vector field vanishes. The first two coefficients in the Taylor expansion of the solution with respect to ϵ are

$$y_1(\theta) = \int_0^{\theta}F_1(\phi,q_0;\sigma_1)\,d\phi$$

and

$$y_2(\theta) = \int_0^{\theta}\left[F_2(\phi,q_0;\sigma_1) + D_qF_1(\phi,q_0;\sigma_1)y_1(\phi)\right]d\phi.$$

Taylor's theorem with parameters gives

$$q(2\pi;q_0,\sigma_1,\epsilon) = q_0 + \epsilon y_1(2\pi) + \epsilon^2 y_2(2\pi) + \epsilon^3\mathcal{R}(q_0,\sigma_1,\epsilon).$$

The definitions of g_1 and g_2 give $y_1(2\pi) = 2\pi g_1(q_0;\sigma_1)$ and $y_2(2\pi) = 2\pi g_2(q_0;\sigma_1)$, which proves the result. □

Let $q_*(\sigma_1)$ be the smooth branch of simple zeros of g_1 obtained near σ_1^*, and set

$$A(\sigma_1) := D_q g_1(q_*(\sigma_1);\sigma_1).$$

Lemma 3.2. *After reducing J_0 and ϵ_0, there exists a unique C^4 branch $(\sigma_1,\epsilon) \mapsto q_\epsilon(\sigma_1)$ of fixed points of Π_{ϵ,σ_1} that continues $q_*(\sigma_1)$. It satisfies, uniformly for $\sigma_1 \in J_0$,*

$$q_\epsilon(\sigma_1) = q_*(\sigma_1) + \epsilon q_1(\sigma_1) + \mathcal{O}(\epsilon^2), \qquad q_1(\sigma_1) = -A(\sigma_1)^{-1} g_2(q_*(\sigma_1);\sigma_1). \tag{62}$$

Proof. For $\epsilon \neq 0$, define

$$\mathcal{F}(q,\sigma_1,\epsilon) = \frac{\Pi_{\epsilon,\sigma_1}(q) - q}{2\pi\epsilon}.$$

Lemma 3.1 provides the C^4 extension $\mathcal{F}(q,\sigma_1,0) = g_1(q;\sigma_1)$, and the expansion $\mathcal{F} = g_1 + \epsilon g_2 + \mathcal{O}(\epsilon^2)$. Since

$$D_q\mathcal{F}(q_*(\sigma_1),\sigma_1,0) = A(\sigma_1)$$

is nonsingular for σ_1 close to σ_1^*, the parameter-dependent implicit-function theorem gives the asserted fixed-point branch and its local uniqueness. Substitution of $q_\epsilon = q_* + \epsilon q_1 + \mathcal{O}(\epsilon^2)$ into $\mathcal{F} = 0$ yields

$$A(\sigma_1)q_1(\sigma_1) + g_2(q_*(\sigma_1);\sigma_1) = 0,$$

which is precisely (62). □

Differentiating (61) and evaluating it at $q_\epsilon(\sigma_1)$ gives

$$D_q\Pi_{\epsilon,\sigma_1}(q_\epsilon(\sigma_1)) = I + 2\pi\epsilon A(\sigma_1) + 2\pi\epsilon^2 K(\sigma_1) + \mathcal{O}(\epsilon^3), \tag{63}$$

where

$$K(\sigma_1) = D_q g_2(q_*(\sigma_1);\sigma_1) + D_q^2 g_1(q_*(\sigma_1);\sigma_1)[q_1(\sigma_1),\cdot]. \tag{64}$$

For the present averaged function,

$$D_q^2 g_1[q_1,\cdot] = \frac{1}{\omega}\begin{pmatrix} B_X q_{1,\zeta} & B_X q_{1,\rho} \\ B_Z q_{1,\rho} & 2B_Z q_{1,\zeta} \end{pmatrix},$$

so K is explicitly determined by (58)–(64). These formulas provide the second-order linear information required to locate the exact unit-modulus curve. They do not, by themselves, establish a Neimark–Sacker bifurcation.

3.4. **The exact unit-modulus curve.** We continue to use σ_1 as the bifurcation parameter and assume the hypotheses of the preceding subsection. The smooth branch $q_*(\sigma_1)$ then satisfies

$$\operatorname{tr} A(\sigma_1) = \tau(\sigma_1), \qquad \tau(\sigma_1^*) = 0, \qquad \tau'(\sigma_1^*) = -\frac{rk_0}{D_0\omega} \neq 0.$$

At the critical value, write

$$q_* := q_*(\sigma_1^*), \qquad A_* := A(\sigma_1^*), \qquad K_* := K(\sigma_1^*).$$

Then

$$\operatorname{tr} A_* = 0, \qquad \det A_* = \nu_0^2, \qquad \nu_0 = \frac{\rho_*\sqrt{-B_X B_Z}}{\omega} > 0.$$

Proposition 3.3. *There exist $\epsilon_1 > 0$, an interval J containing σ_1^*, and a unique C^2 function*

$$\sigma_c : (-\epsilon_1,\epsilon_1) \to J, \qquad \sigma_c(0) = \sigma_1^*,$$

such that, for every $0 < |\epsilon| < \epsilon_1$, the two transverse Floquet multipliers of the periodic orbit corresponding to $q_\epsilon(\sigma_c(\epsilon))$ form a nonreal conjugate pair of unit modulus. No other $\sigma_1 \in J$ on this local periodic-orbit branch has this property. Moreover,

$$\sigma_c(\epsilon) = \frac{2dx_0}{k_0}\beta_1 + \epsilon\frac{D_0\omega}{rk_0}\big(\operatorname{tr} K_* + 2\pi\nu_0^2\big) + \mathcal{O}(\epsilon^2), \tag{65}$$

and the unit-circle crossing is transverse in the sense that

$$\left.\frac{\partial|\mu_\pm(\sigma_1,\epsilon)|}{\partial\sigma_1}\right|_{\sigma_1=\sigma_c(\epsilon)} = \pi\epsilon\tau'(\sigma_1^*) + \mathcal{O}(\epsilon^2) = -\frac{\pi r k_0}{D_0\omega}\epsilon + \mathcal{O}(\epsilon^2) \neq 0. \tag{66}$$

Proof. For σ_1 near σ_1^*, set

$$M(\sigma_1,\epsilon) := D_q\Pi_{\epsilon,\sigma_1}\big(q_\epsilon(\sigma_1)\big).$$

For $\epsilon \neq 0$, define

$$\Phi(\sigma_1,\epsilon) := \frac{\det M(\sigma_1,\epsilon) - 1}{2\pi\epsilon}.$$

By (63), for 2×2 matrices,

$$\det M(\sigma_1,\epsilon) = 1 + 2\pi\epsilon\ \operatorname{tr} A(\sigma_1) + 2\pi\epsilon^2\big[\operatorname{tr} K(\sigma_1) + 2\pi\det A(\sigma_1)\big] + \mathcal{O}(\epsilon^3). \tag{67}$$

Hence Φ admits the C^2 extension

$$\Phi(\sigma_1,\epsilon) = \tau(\sigma_1) + \epsilon\big[\operatorname{tr} K(\sigma_1) + 2\pi\det A(\sigma_1)\big] + \mathcal{O}(\epsilon^2), \tag{68}$$

where

$$\Phi(\sigma_1,0) := \tau(\sigma_1).$$

At $(\sigma_1^*,0)$,

$$\Phi(\sigma_1^*,0) = 0, \qquad \partial_{\sigma_1}\Phi(\sigma_1^*,0) = \tau'(\sigma_1^*) \neq 0.$$

The implicit-function theorem therefore yields an interval J, a number $\epsilon_1 > 0$, and a unique C^2 function $\sigma_c : (-\epsilon_1,\epsilon_1) \to J$ satisfying

$$\Phi(\sigma_c(\epsilon),\epsilon) = 0, \qquad \sigma_c(0) = \sigma_1^*.$$

For $\epsilon \neq 0$, this is equivalent to

$$\det M(\sigma_c(\epsilon),\epsilon) = 1.$$

Differentiating

$$\Phi(\sigma_c(\epsilon),\epsilon) = 0$$

with respect to ϵ at $\epsilon = 0$ gives

$$\tau'(\sigma_1^*)\sigma_c'(0) + \operatorname{tr} K_* + 2\pi\det A_* = 0.$$

Therefore,

$$\sigma_c'(0) = -\frac{\operatorname{tr} K_* + 2\pi\det A_*}{\tau'(\sigma_1^*)}.$$

Using

$$\det A_* = \nu_0^2, \qquad \tau'(\sigma_1^*) = -\frac{rk_0}{D_0\omega},$$

we obtain

$$\sigma_c'(0) = \frac{D_0\omega}{rk_0}\big(\operatorname{tr} K_* + 2\pi\nu_0^2\big),$$

which proves (65).

It remains to verify that the corresponding multipliers form a nonreal conjugate pair. Since

$$\operatorname{tr} A_* = 0, \qquad \det A_* = \nu_0^2 > 0, \qquad \sigma_c(\epsilon) = \sigma_1^* + \mathcal{O}(\epsilon),$$

we have, with

$$M_\epsilon := M(\sigma_c(\epsilon),\epsilon),$$

the discriminant expansion

$$\left(\operatorname{tr} M_\epsilon\right)^2 - 4\det M_\epsilon = -16\pi^2\epsilon^2\nu_0^2 + \mathcal{O}(\epsilon^3) < 0$$

for every sufficiently small $\epsilon \neq 0$. Thus the two eigenvalues of M_ϵ form a nonreal conjugate pair. Since

$$\det M_\epsilon = 1,$$

their product is one, and hence both multipliers have unit modulus.

After reducing J if necessary, continuity implies that the discriminant remains negative for every $\sigma_1 \in J$ and all sufficiently small $\epsilon \neq 0$. Thus the transverse multipliers remain a nonreal conjugate pair throughout this neighborhood, and

$$|\mu_\pm(\sigma_1, \epsilon)|^2 = \det M(\sigma_1, \epsilon).$$

Consequently, they have unit modulus if and only if

$$\det M(\sigma_1, \epsilon) = 1,$$

which, for $\epsilon \neq 0$, is equivalent to

$$\Phi(\sigma_1, \epsilon) = 0.$$

By the local uniqueness supplied by the implicit-function theorem, this occurs precisely at

$$\sigma_1 = \sigma_c(\epsilon).$$

Finally, differentiating (67) with respect to σ_1 and evaluating at $\sigma_1 = \sigma_c(\epsilon)$ gives

$$\frac{\partial}{\partial\sigma_1}\det M(\sigma_1, \epsilon)\Big|_{\sigma_1=\sigma_c(\epsilon)} = 2\pi\epsilon\tau'(\sigma_1^*) + \mathcal{O}(\epsilon^2).$$

Since the multipliers are nonreal conjugates in J,

$$|\mu_\pm|^2 = \det M.$$

On the critical curve $|\mu_\pm| = 1$, and therefore

$$\frac{\partial|\mu_\pm(\sigma_1, \epsilon)|}{\partial\sigma_1}\Big|_{\sigma_1=\sigma_c(\epsilon)} = \frac{1}{2}\,\frac{\partial}{\partial\sigma_1}\det M(\sigma_1, \epsilon)\Big|_{\sigma_1=\sigma_c(\epsilon)}.$$

Hence

$$\frac{\partial|\mu_\pm(\sigma_1, \epsilon)|}{\partial\sigma_1}\Big|_{\sigma_1=\sigma_c(\epsilon)} = \pi\epsilon\tau'(\sigma_1^*) + \mathcal{O}(\epsilon^2) = -\frac{\pi r k_0}{D_0\omega}\epsilon + \mathcal{O}(\epsilon^2) \neq 0$$

for every sufficiently small $\epsilon \neq 0$. This proves the transverse crossing. □

Remark 3.4. Proposition 3.3 establishes an exact transversal unit-modulus crossing, but it does not yet imply the existence of an invariant closed curve. A Neimark–Sacker bifurcation requires, in addition, the low-order nonresonance conditions and a nonzero first Lyapunov coefficient for the centered Poincaré map.

3.5. **Centered Poincaré map and cubic expansion.** Throughout this subsection we impose the standing assumptions

$$2dx_0(1-r) > 1, \qquad k_0 D_0 \neq 0, \qquad B_X B_Z < 0, \qquad A_X(\sigma_1^*) \neq 0, \tag{69}$$

and use the exact quadratic unfolding (12). Under these assumptions, Proposition 3.3 provides the critical curve $\sigma_1 = \sigma_c(\epsilon)$, and Lemma 3.1 provides the required joint C^4 expansion of the return map.

Let

$$\widehat{q_\epsilon} := q_\epsilon(\sigma_c(\epsilon)).$$

If $\sigma_{1,1} := \sigma_c'(0)$, then

$$\widehat{q_\epsilon} = q_* + \epsilon\widehat{q_1} + \mathcal{O}(\epsilon^2), \qquad \widehat{q_1} = \sigma_{1,1}q_*'(\sigma_1^*) + q_{1,*}, \qquad q_{1,*} = -A_*^{-1}g_2(q_*; \sigma_1^*). \tag{70}$$

Center the Poincaré map at this fixed point:

$$H_\epsilon(x) := \Pi_{\epsilon,\sigma_c(\epsilon)}(\widehat{q}_\epsilon + x) - \widehat{q}_\epsilon.$$

Then $H_\epsilon(0) = 0$. Taylor's theorem gives the exact decomposition

$$H_\epsilon(x) = L_\epsilon x + \frac{1}{2}B_\epsilon(x,x) + \frac{1}{6}C_\epsilon(x,x,x) + R_{4,\epsilon}(x), \tag{71}$$

where $L_\epsilon, B_\epsilon, C_\epsilon$ are, respectively, the first, second, and third state derivatives of $\Pi_{\epsilon,\sigma_c(\epsilon)}$ at $\widehat{q}_\epsilon$. Moreover, there are constants $C > 0, \delta > 0, \epsilon_0 > 0$ such that

$$\|R_{4,\epsilon}(x)\| \le C\|x\|^4 \qquad (0 < |\epsilon| < \epsilon_0,\ \|x\| < \delta).$$

Thus (71) is an exact Taylor identity with a quantified remainder; the subsequent formulas in powers of ϵ are asymptotic expansions.

From (63) and the expansion of $\sigma_c(\epsilon)$,

$$L_\epsilon = I + 2\pi\epsilon A_* + 2\pi\epsilon^2\big(\sigma_{1,1}A'_* + K_*\big) + \mathcal{O}(\epsilon^3), \tag{72}$$

where

$$A'_* := \frac{d}{d\sigma_1}A(\sigma_1)\Big|_{\sigma_1=\sigma_1^*}.$$

Equivalently, the second-order linear coefficient can be written as

$$A_2 = D_q^2 g_1[\widehat{q}_1, \cdot] + \sigma_{1,1}\partial_{\sigma_1}D_q g_1 + D_q g_2, \tag{73}$$

where all quantities on the right-hand side are evaluated at (q_*, σ_1^*). Indeed,

$$A_2 = \sigma_{1,1}A'_* + K_*.$$

At $\sigma_1 = \sigma_1^*$, the trace condition gives $A_Z + 2B_Z\zeta_* = 0$. Hence

$$A_* = \frac{\rho_*}{\omega}\begin{pmatrix} 0 & B_X \\ B_Z & 0 \end{pmatrix}, \qquad A_*^2 = -\nu_0^2 I, \qquad \nu_0^2 = -\frac{B_X B_Z \rho_*^2}{\omega^2} > 0. \tag{74}$$

By Proposition 3.3, $\det L_\epsilon = 1$, and the eigenvalues of L_ϵ form a nonreal conjugate pair. Thus they can be written as $e^{\pm i\vartheta_\epsilon}$, where the signed critical angle is chosen so that

$$\vartheta_\epsilon = 2\pi\epsilon\nu_0 + \mathcal{O}(\epsilon^2). \tag{75}$$

In particular, $\vartheta_\epsilon \neq 0$ for every sufficiently small $\epsilon \neq 0$.

Since g_1 is quadratic in q, $D_q^3 g_1 \equiv 0$, while $D_q^2 g_1$ is constant in q. Moreover, B_X and B_Z are independent of σ_1, so $\partial_{\sigma_1}D_q^2 g_1 = 0$. Differentiating the return-map expansion and evaluating it at $(\widehat{q}_\epsilon, \sigma_c(\epsilon))$ therefore gives

$$\begin{aligned} &B_\epsilon = 2\pi\epsilon B_1 + 2\pi\epsilon^2 B_2 + \mathcal{O}(\epsilon^3), \quad B_1 = D_q^2 g_1, \qquad\qquad B_2 = D_q^2 g_2(q_*; \sigma_1^*), \\ &C_\epsilon = 2\pi\epsilon^2 C_2 + \mathcal{O}(\epsilon^3), \qquad\qquad C_2 = D_q^3 g_2(q_*; \sigma_1^*). \end{aligned} \tag{76}$$

No term involving $\widehat{q}_1$ appears at the displayed orders because $D_q^3 g_1 = 0$ and $\partial_{\sigma_1}D_q^2 g_1 = 0$. Explicitly, for $u = (u_1, u_2)^\mathsf{T}$ and $v = (v_1, v_2)^\mathsf{T}$,

$$B_1(u,v) = \frac{1}{\omega}\begin{pmatrix} B_X(u_1 v_2 + u_2 v_1) \\ B_Z u_1 v_1 + 2B_Z u_2 v_2 \end{pmatrix}.$$

The tensors B_2 and C_2 are obtained by differentiating the explicit polynomial g_2 in (58)–(60).

We next place the linear part in rotation form. Write

$$L_\epsilon = \begin{pmatrix} a_\epsilon & b_\epsilon \\ c_\epsilon & d_\epsilon \end{pmatrix}.$$

Equations (72) and (74) imply

$$b_\epsilon = \frac{2\pi\epsilon B_X \rho_*}{\omega} + \mathcal{O}(\epsilon^2) \neq 0.$$

Define

$$S_\epsilon = \begin{pmatrix} 1 & 0 \\ \dfrac{\cos\vartheta_\epsilon - a_\epsilon}{b_\epsilon} & -\dfrac{\sin\vartheta_\epsilon}{b_\epsilon} \end{pmatrix}.$$

Since $\sin\vartheta_\epsilon = 2\pi\epsilon\nu_0 + \mathcal{O}(\epsilon^2)$, both S_ϵ and S_ϵ^{-1} remain bounded as $\epsilon \to 0$, and

$$S_\epsilon^{-1} L_\epsilon S_\epsilon = R_{\vartheta_\epsilon} := \begin{pmatrix} \cos\vartheta_\epsilon & -\sin\vartheta_\epsilon \\ \sin\vartheta_\epsilon & \cos\vartheta_\epsilon \end{pmatrix}. \tag{77}$$

Under the change of variables $x = S_\epsilon y$, the centered map has the exact third-order Taylor decomposition

$$\widetilde{H}_\epsilon(y) = R_{\vartheta_\epsilon} y + \frac{1}{2}\widetilde{B}_\epsilon(y,y) + \frac{1}{6}\widetilde{C}_\epsilon(y,y,y) + \widetilde{R}_{4,\epsilon}(y), \tag{78}$$

where

$$\begin{aligned} \widetilde{B}_\epsilon(u,v) &= S_\epsilon^{-1} B_\epsilon(S_\epsilon u, S_\epsilon v), \\ \widetilde{C}_\epsilon(u,v,w) &= S_\epsilon^{-1} C_\epsilon(S_\epsilon u, S_\epsilon v, S_\epsilon w), \\ \widetilde{R}_{4,\epsilon}(y) &= S_\epsilon^{-1} R_{4,\epsilon}(S_\epsilon y). \end{aligned}$$

Because S_ϵ and S_ϵ^{-1} are uniformly bounded, there is a constant $\widetilde{C} > 0$, independent of sufficiently small $\epsilon \neq 0$, such that

$$\|\widetilde{R}_{4,\epsilon}(y)\| \le \widetilde{C}\|y\|^4.$$

Proposition 3.5. *Assume* (69) *and the exact quadratic unfolding* (12)*. For every sufficiently small $\epsilon \neq 0$, the Poincaré map at $\sigma_1 = \sigma_c(\epsilon)$, centered at $\widehat{q}_\epsilon$, is linearly conjugate to the map* (78)*. Its linear part is the rotation R_{ϑ_ϵ}, where*

$$\vartheta_\epsilon = 2\pi\epsilon\nu_0 + \mathcal{O}(\epsilon^2),$$

and its quadratic and cubic tensors are determined by (76) *and* (77)*.*

Remark 3.6. The identities in this subsection establish the cubic Taylor jet of the centered Poincaré map. They do not, by themselves, determine the first Neimark–Sacker Lyapunov coefficient. Since $\vartheta_\epsilon \to 0$, the relevant homological resolvents are $\mathcal{O}(|\epsilon|^{-1})$; consequently, the next subsection must substitute the exact tensors into the Neimark–Sacker coefficient formula and retain their full ϵ-dependence. Nonresonance is pointwise in ϵ, not uniform as $\epsilon \to 0$.

3.6. **First Lyapunov coefficient.** Appendix A carries out the exact symbolic coefficient extraction used below and forms part of the proof of Lemma 3.7.

We retain the standing hypotheses (69), use the exact quadratic unfolding (12), and restrict the parameter to the exact unit-modulus curve $\sigma_1 = \sigma_c(\epsilon)$. By Proposition 3.5, for every sufficiently small $\epsilon \neq 0$, the centered Poincaré map in rotation coordinates has the cubic Taylor decomposition

$$\widetilde{H}_\epsilon(y) = R_{\vartheta_\epsilon} y + \frac{1}{2}\widetilde{B}_\epsilon(y,y) + \frac{1}{6}\widetilde{C}_\epsilon(y,y,y) + \widetilde{R}_{4,\epsilon}(y), \qquad \|\widetilde{R}_{4,\epsilon}(y)\| \le C\|y\|^4,$$

where

$$R_{\vartheta_\epsilon} = \begin{pmatrix} \cos\vartheta_\epsilon & -\sin\vartheta_\epsilon \\ \sin\vartheta_\epsilon & \cos\vartheta_\epsilon \end{pmatrix}, \qquad \vartheta_\epsilon = 2\pi\epsilon\nu_0 + \mathcal{O}(\epsilon^2).$$

Set $\lambda_\epsilon = e^{i\vartheta_\epsilon}$. Since $\nu_0 > 0$, there exists $\epsilon_0 > 0$ such that

$$\lambda_\epsilon^j \neq 1, \qquad j = 1,2,3,4, \qquad 0 < |\epsilon| < \epsilon_0. \tag{79}$$

Thus the homological operators used below are invertible for every fixed ϵ in this range.

Let

$$q = \begin{pmatrix} 1 \\ -i \end{pmatrix}, \qquad p_{\mathrm{ad}} = \frac{1}{2}\begin{pmatrix} 1 \\ -i \end{pmatrix}, \qquad \langle u, v\rangle = \overline{u}^{\mathsf{T}} v.$$

Then

$$R_{\vartheta_\epsilon} q = \lambda_\epsilon q, \qquad R_{\vartheta_\epsilon}^{\mathsf{T}} p_{\mathrm{ad}} = \overline{\lambda_\epsilon} p_{\mathrm{ad}}, \qquad \langle p_{\mathrm{ad}}, q\rangle = 1.$$

Define

$$h_{11} = (I - R_{\vartheta_\epsilon})^{-1} \widetilde{B}_\epsilon(q, \overline{q}),$$
$$h_{20} = (\lambda_\epsilon^2 I - R_{\vartheta_\epsilon})^{-1} \widetilde{B}_\epsilon(q, q).$$

With this normalization, the first Neimark–Sacker Lyapunov coefficient is

$$l_1^\epsilon = \frac{1}{2}\,\mathrm{Re}\left[e^{-i\vartheta_\epsilon}\left\langle p_{\mathrm{ad}}, \widetilde{C}_\epsilon(q, q, \overline{q}) + 2\widetilde{B}_\epsilon(q, h_{11}) + \widetilde{B}_\epsilon(\overline{q}, h_{20})\right\rangle\right]. \tag{80}$$

Lemma 3.7. *Along the exact unit-modulus curve $\sigma_1 = \sigma_c(\epsilon)$, the coefficient* (80) *satisfies*

$$l_1^\epsilon = \epsilon^2 L_{\mathrm{HR}} + \mathcal{O}(\epsilon^3), \tag{81}$$

where

$$L_{\mathrm{HR}} = -\frac{\pi B_Z(B_X + B_Z)}{2B_X^2\omega^2}\big(2B_X B_Y - 7B_Y B_Z - 3E_X\omega\big). \tag{82}$$

Consequently,

$$\lim_{\epsilon\to 0} \frac{l_1^\epsilon}{\epsilon^2} = L_{\mathrm{HR}}.$$

Proof. Appendix A gives the complete coefficient extraction. The calculation retains the displacement of the critical fixed point, the correction to the exact unit-modulus curve, the expansion of the real rotation conjugacy, and the singular homological resolvents. It gives a vanishing coefficient of ϵ and the coefficient in (82) at order ϵ^2. The remainder analysis in the appendix shows that every omitted term in the normal-form coefficient contributes only $\mathcal{O}(\epsilon^3)$. □

Corollary 3.8. *The coefficient in* (82) *has the equivalent Hindmarsh–Rose representation*

$$L_{\mathrm{HR}} = -\frac{\pi k_0 r(k_0 - d)}{2\omega^3 D_0^2}\, P_{\mathrm{HR}}. \tag{83}$$

Proof. Substitute

$$B_X = \frac{D_0}{\omega^2}, \qquad B_Y = \frac{d - (1+r)k_0}{\omega}, \qquad B_Z = \frac{r(k_0 - d)}{\omega^2}, \qquad E_X = -\frac{a(\omega^2 - r)}{\omega^2}$$

into (82). The identity

$$D_0 + r(k_0 - d) = k_0\omega^2$$

then gives (83) by direct simplification. □

Proposition 3.9. *Under* (69) *and the exact quadratic unfolding* (12)*, the first Neimark–Sacker Lyapunov coefficient on the exact unit-modulus curve satisfies* (81)*, and*

$$L_{\mathrm{HR}} \neq 0 \quad \Longleftrightarrow \quad P_{\mathrm{HR}} \neq 0.$$

More precisely, if $P_{\mathrm{HR}} \neq 0$, then there exists $\epsilon_1 > 0$ such that

$$l_1^\epsilon \neq 0, \qquad \operatorname{sgn} l_1^\epsilon = \operatorname{sgn} L_{\mathrm{HR}}, \qquad 0 < |\epsilon| < \epsilon_1.$$

Proof. By Corollary 3.8,

$$L_{\mathrm{HR}} = -\frac{\pi k_0 r(k_0 - d)}{2\omega^3 D_0^2}\, P_{\mathrm{HR}}.$$

The standing hypotheses imply

$$\omega > 0, \qquad r > 0, \qquad k_0 D_0 \neq 0, \qquad B_X B_Z < 0.$$

Since

$$B_X B_Z = \frac{r D_0(k_0 - d)}{\omega^4},$$

they also imply $k_0 - d \neq 0$. Hence the prefactor multiplying P_{HR} is nonzero, which proves the first assertion. If $P_{\mathrm{HR}} \neq 0$, (81) gives

$$\frac{l_1^\epsilon}{\epsilon^2} = L_{\mathrm{HR}} + \mathcal{O}(\epsilon) \longrightarrow L_{\mathrm{HR}},$$

and the stated nonvanishing and sign conclusion follows. □

Remark 3.10. The nonresonance condition (79) is pointwise in ϵ, not uniform as $\epsilon \to 0$, because $\lambda_\epsilon \to 1$. The coefficient L_{HR} is independent of β_2 and σ_2. These parameters affect the order-ϵ corrections of the critical fixed point and the unit-modulus curve, but their contributions cancel from the first nonzero term of l_1^ϵ.

Lemma 3.11. *Assume*

$$D_0(k_0 - d) < 0, \qquad P_{\mathrm{HR}} \neq 0,$$

and let

$$\alpha_\epsilon := \left.\frac{\partial|\mu_+(\sigma_1,\epsilon)|}{\partial\sigma_1}\right|_{\sigma_1=\sigma_c(\epsilon)}, \qquad b_\epsilon := l_1^\epsilon.$$

Then there exists $\epsilon_1 > 0$ such that, for every $0 < \epsilon < \epsilon_1$, $\alpha_\epsilon b_\epsilon \neq 0$ and

$$\alpha_\epsilon b_\epsilon\big(\sigma_1 - \sigma_c(\epsilon)\big) < 0 \quad \Longleftrightarrow \quad P_{\mathrm{HR}}\big(\sigma_1 - \sigma_c(\epsilon)\big) > 0. \tag{84}$$

Proof. Proposition 3.3 gives

$$\alpha_\epsilon = \pi\epsilon\tau'(\sigma_1^*) + \mathcal{O}(\epsilon^2) = -\frac{\pi r k_0}{D_0\omega}\epsilon + \mathcal{O}(\epsilon^2).$$

On the other hand, Proposition 3.9 and Corollary 3.8 give

$$b_\epsilon = -\frac{\pi k_0 r(k_0 - d)}{2\omega^3 D_0^2} P_{\mathrm{HR}}\epsilon^2 + \mathcal{O}(\epsilon^3).$$

Multiplication yields

$$\alpha_\epsilon b_\epsilon = \frac{\pi^2 r^2 k_0^2 (k_0 - d)}{2\omega^4 D_0^3} P_{\mathrm{HR}}\epsilon^3 + \mathcal{O}(\epsilon^4).$$

Because $D_0(k_0 - d) < 0$, the leading prefactor has sign $-\operatorname{sgn} P_{\mathrm{HR}}$. Hence, for every sufficiently small $\epsilon > 0$,

$$\operatorname{sgn}(\alpha_\epsilon b_\epsilon) = -\operatorname{sgn} P_{\mathrm{HR}},$$

which proves (84). □

3.7. **An abstract Neimark–Sacker theorem.** We use the following standard local Neimark–Sacker theorem in the normalization of (80); see, for example, [15, Chapter 4].

Theorem 3.12 (Local Neimark–Sacker bifurcation)**.** *Let*

$$F_\eta : \mathbb{R}^2 \longrightarrow \mathbb{R}^2, \qquad |\eta| < \eta_0,$$

be a C^4 one-parameter family of local diffeomorphisms satisfying $F_\eta(0) = 0$. Suppose that the eigenvalues of $DF_0(0)$ are

$$\lambda_0 = e^{i\vartheta_0}, \qquad \overline{\lambda_0} = e^{-i\vartheta_0}, \qquad 0 < \vartheta_0 < \pi,$$

and that the low-order nonresonance conditions

$$\lambda_0^j \neq 1, \qquad j = 1, 2, 3, 4, \tag{85}$$

hold. Let $\lambda(\eta)$ be the continuation of λ_0, chosen with positive imaginary part, and assume the transverse crossing condition

$$\left.\frac{d|\lambda(\eta)|}{d\eta}\right|_{\eta=0} \neq 0. \tag{86}$$

Write the cubic Taylor expansion of F_0 as

$$F_0(x) = Ax + \frac{1}{2}B(x,x) + \frac{1}{6}C(x,x,x) + \mathcal{O}(\|x\|^4), \qquad A := DF_0(0).$$

Choose $q, p_{\mathrm{ad}} \in \mathbb{C}^2$ so that

$$Aq = \lambda_0 q, \qquad A^{\mathsf{T}} p_{\mathrm{ad}} = \overline{\lambda_0} p_{\mathrm{ad}}, \qquad \langle p_{\mathrm{ad}}, q\rangle = 1.$$

Define

$$h_{11} = (I-A)^{-1}B(q,\overline{q}), \qquad h_{20} = (\lambda_0^2 I - A)^{-1}B(q,q),$$

and let

$$b_{\mathrm{NS}} = \frac{1}{2}\,\mathrm{Re}\left[\lambda_0^{-1}\,\langle p_{\mathrm{ad}}, C(q,q,\overline{q}) + 2B(q,h_{11}) + B(\overline{q},h_{20})\rangle\right].$$

Assume $b_{\mathrm{NS}} \neq 0$. Then there exist $\eta_1 > 0$ and a neighborhood U of the origin such that:

1. *a unique local branch of invariant closed curves bifurcates from the fixed point $x = 0$;*
2. *after reducing η_1 and U, this branch exists precisely on the side*

$$a\, b_{\mathrm{NS}}\eta < 0, \qquad a := \left.\frac{d|\lambda(\eta)|}{d\eta}\right|_{\eta=0};$$

3. *if $b_{\mathrm{NS}} < 0$, the bifurcation is supercritical and the bifurcating invariant closed curve is attracting, while the fixed point on its existence side is unstable;*
4. *if $b_{\mathrm{NS}} > 0$, the bifurcation is subcritical and the bifurcating invariant closed curve is repelling, while the fixed point on its existence side is asymptotically stable.*

Remark 3.13. The uniqueness asserted in Theorem 3.12 concerns the local branch of invariant closed curves bifurcating from the critical fixed point. No global uniqueness of invariant closed curves is asserted.

3.8. **Proof of Theorem 1.3.** Fix $0 < \epsilon < \epsilon_0$, reducing ϵ_0 whenever necessary. The standing hypotheses imply

$$B_X B_Z < 0, \qquad A_X(\sigma_1^*) \neq 0.$$

Consequently, the nontrivial zero q_* of the first averaged function is simple, since

$$\det D_q g_1(q_*;\sigma_1^*) = -\frac{B_X B_Z \rho_*^2}{\omega^2} > 0.$$

Lemma 3.2 therefore provides a unique local branch $q_\epsilon(\sigma_1)$ of fixed points of the Poincaré map near (q_*,σ_1^*). These fixed points correspond to the local family of periodic orbits $\Gamma_{\sigma_1,\epsilon}$. Proposition 3.3 provides a unique critical parameter $\sigma_c(\epsilon)$ near σ_1^* such that

$$\det D_q \Pi_{\epsilon,\sigma_c(\epsilon)}\big(q_\epsilon(\sigma_c(\epsilon))\big) = 1.$$

At this parameter, the discriminant of the transverse linearization is negative. Hence its two eigenvalues form a nonreal conjugate pair. Their product is one, and therefore both multipliers have unit modulus. The same proposition gives the modulus-crossing coefficient

$$\begin{aligned}\alpha_\epsilon &:= \left.\frac{\partial|\mu_+(\sigma_1,\epsilon)|}{\partial\sigma_1}\right|_{\sigma_1=\sigma_c(\epsilon)} \\ &= \pi\epsilon\tau'(\sigma_1^*) + \mathcal{O}(\epsilon^2) = -\frac{\pi r k_0}{D_0\omega}\epsilon + \mathcal{O}(\epsilon^2).\end{aligned} \tag{87}$$

Since $rk_0/(D_0\omega) \neq 0$, one has $\alpha_\epsilon \neq 0$ for every sufficiently small $\epsilon > 0$.

For fixed $\epsilon > 0$, introduce the local parameter

$$\eta := \sigma_1 - \sigma_c(\epsilon)$$

and define the centered family of Poincaré maps

$$F_{\epsilon,\eta}(x) := \Pi_{\epsilon,\sigma_c(\epsilon)+\eta}\big(q_\epsilon(\sigma_c(\epsilon)+\eta)+x\big) - q_\epsilon(\sigma_c(\epsilon)+\eta).$$

By construction, $F_{\epsilon,\eta}(0) = 0$. For fixed $\epsilon \neq 0$, the Poincaré map is C^4 jointly in its state variable and in σ_1. Moreover, the parameter-dependent implicit-function theorem applied to the fixed-point equation shows that $\sigma_1 \mapsto q_\epsilon(\sigma_1)$ is C^4. Consequently, $(\eta, x) \mapsto F_{\epsilon,\eta}(x)$ is a C^4 family of local diffeomorphisms, as required in Theorem 3.12.

At $\eta = 0$, the derivative

$$D_x F_{\epsilon,0}(0) = D_q \Pi_{\epsilon,\sigma_c(\epsilon)}\big(q_\epsilon(\sigma_c(\epsilon))\big)$$

has the nonreal conjugate eigenvalue pair $e^{\pm i\vartheta_\epsilon}$. By Proposition 3.5,

$$\vartheta_\epsilon = 2\pi\epsilon\nu_0 + \mathcal{O}(\epsilon^2), \qquad \nu_0 > 0.$$

After reducing ϵ_0, we may assume $0 < |\vartheta_\epsilon| < \pi/2$. If $e^{ij\vartheta_\epsilon} = 1$ for some $j \in \{1,2,3,4\}$, then $j\vartheta_\epsilon = 2\pi m$ for some $m \in \mathbb{Z}$. However, $0 < |j\vartheta_\epsilon| < 2\pi$, which is impossible. Hence

$$e^{ij\vartheta_\epsilon} \neq 1, \qquad j = 1,2,3,4.$$

Thus the low-order nonresonance hypothesis (85) is satisfied.

The modulus derivative of the critical multiplier of $F_{\epsilon,\eta}$ at $\eta = 0$ is precisely α_ϵ from (87). This verifies (86). Let $b_\epsilon := l_1^\epsilon$ be the first Neimark–Sacker Lyapunov coefficient in the normalization (80). By Lemma 3.7 and Corollary 3.8,

$$b_\epsilon = \epsilon^2 L_{\mathrm{HR}} + \mathcal{O}(\epsilon^3), \qquad L_{\mathrm{HR}} = -\frac{\pi k_0 r(k_0 - d)}{2\omega^3 D_0^2} P_{\mathrm{HR}}.$$

The standing hypotheses imply $b_\epsilon \neq 0$ for every sufficiently small $\epsilon > 0$. All hypotheses of Theorem 3.12 have now been verified for the family $F_{\epsilon,\eta}$, with

$$a = \alpha_\epsilon, \qquad b_{\mathrm{NS}} = b_\epsilon, \qquad \eta = \sigma_1 - \sigma_c(\epsilon).$$

Consequently, after reducing an interval J_ϵ containing $\sigma_c(\epsilon)$ and a neighborhood V_ϵ of the critical fixed point in the local transverse section, a unique local branch of invariant closed curves bifurcates from the critical fixed point. By Theorem 3.12, this branch exists precisely when

$$\alpha_\epsilon b_\epsilon\big(\sigma_1 - \sigma_c(\epsilon)\big) < 0.$$

Lemma 3.11 gives the equivalent condition

$$P_{\mathrm{HR}}\big(\sigma_1 - \sigma_c(\epsilon)\big) > 0.$$

No member of this local bifurcating branch lies in V_ϵ on the opposite side of the critical parameter.

It remains to determine criticality and stability. If $L_{\mathrm{HR}} < 0$, then $b_\epsilon < 0$ and Theorem 3.12 shows that the bifurcation is supercritical. If $L_{\mathrm{HR}} > 0$, then $b_\epsilon > 0$ and the bifurcation is subcritical. Since fixed points of the Poincaré map correspond to periodic orbits of the flow, these conclusions give the transverse stability properties of $\Gamma_{\sigma_1,\epsilon}$ stated in Theorem 1.3.

Finally, let $C_{\sigma_1,\epsilon} \subset V_\epsilon$ be a bifurcating invariant closed curve. Choose the local return section and flow tube small enough that its flow segments do not meet before the first return. The restriction of the Poincaré map to $C_{\sigma_1,\epsilon}$ is a circle diffeomorphism of degree $+1$; it is isotopic, through local diffeomorphisms, to the linear rotation R_{ϑ_ϵ}. Its flow suspension is therefore the mapping torus of an orientation-preserving circle diffeomorphism, hence an embedded invariant manifold diffeomorphic to $\mathbb{T}^2$. Normal attraction or repulsion of the invariant closed curve is preserved under suspension. This proves Theorem 1.3.

Remark 3.14. The interval J_ϵ and the neighborhood of the resulting torus may depend on ϵ. No neighborhood uniform in ϵ is asserted, because the critical multipliers converge to 1 as $\epsilon \to 0$.

Remark 3.15. If $P_{\mathrm{HR}} = 0$, then $L_{\mathrm{HR}} = 0$, and the order-ϵ^2 term of l_1^ϵ vanishes. The present calculation does not determine whether the exact coefficient l_1^ϵ vanishes for fixed ϵ, nor does it classify the resulting bifurcation. That case requires the next nonzero term in the Lyapunov-coefficient expansion and, if necessary, higher-order averaging and normal-form calculations.

Example 3.16 (A nonempty algebraic parameter regime). Take

$$a = \frac{1}{2}, \quad c = d = 1, \quad r = 2, \quad x_R = -\frac{8}{5}, \quad x_0 = -1, \quad \beta_1 = 1, \quad \beta_2 = \sigma_2 = 0.$$

Then

$$\omega = 1, \qquad b_0 = -\frac{9}{4}, \quad s_0 = 5, \quad I_0 = \frac{19}{4}, \quad k_0 = -\frac{3}{4}, \quad D_0 = \frac{11}{4}, \quad \sigma_1^* = \frac{8}{3}.$$

At the leading critical value,

$$B_X = \frac{11}{4}, \qquad B_Z = -\frac{7}{2}, \qquad A_X(\sigma_1^*) = \frac{22}{3},$$

so that $B_X B_Z < 0$ and $A_X(\sigma_1^*) \neq 0$. Moreover,

$$P_{\mathrm{HR}} = 96, \qquad L_{\mathrm{HR}} = -\frac{2016\pi}{121} < 0.$$

Thus all nondegeneracy conditions in Theorem 1.3 are satisfied. Evaluating the first correction in Proposition 3.3 gives

$$\sigma_c(\epsilon) = \frac{8}{3} + \frac{76}{9}\epsilon + \mathcal{O}(\epsilon^2). \tag{88}$$

Since $P_{\mathrm{HR}} > 0$, the local torus branch lies on the side $\sigma_1 > \sigma_c(\epsilon)$; since $L_{\mathrm{HR}} < 0$, the bifurcation is supercritical and the bifurcating torus is normally attracting.

This example is used as a mathematical validation regime. In accordance with Remark 1.1, it is not in the conventional slow-adaptation HR parameter regime.

4. Full-system numerical validation

4.1. **Critical curve and Floquet crossing.** The analytical results above do not depend on numerical continuation. We nevertheless use the full unfolded Hindmarsh–Rose system to test two quantitative predictions: the transverse unit-modulus crossing of the nontrivial Floquet pair and the asymptotic displacement of the corresponding critical parameter. We use the algebraic parameter regime of Example 3.16, retain $\beta_2 = \sigma_2 = 0$, and compute the small periodic orbit by single shooting with the phase condition $x(0) = x_0$. The full variational equation is integrated over the numerically determined period, and the numerical critical parameter $\widehat{\sigma}_c(\epsilon)$ is defined by the crossing condition

$$|\mu_\pm(\widehat{\sigma}_c(\epsilon), \epsilon)| = 1.$$

For this parameter set, Proposition 3.3 and (88) predict

$$\sigma_c(\epsilon) = \sigma_1^* + \sigma_{1,1}\epsilon + \mathcal{O}(\epsilon^2), \qquad \sigma_1^* = \frac{8}{3}, \qquad \sigma_{1,1} = \frac{76}{9}.$$

At $\epsilon = 10^{-2}$, the leading prediction is $\sigma_1^* = 2.666666\ldots$, whereas inclusion of the first correction gives

$$\sigma_1^* + \frac{76}{9}\epsilon = 2.751111\ldots.$$

The full-system Floquet calculation gives

$$\widehat{\sigma}_c(10^{-2}) = 2.758704\ldots.$$

TABLE 1. Convergence of the full-system Floquet crossing to the asymptotic critical curve in Example 3.16. The final column should converge to $\sigma_{1,1} = 76/9 = 8.44444\ldots$ as $\epsilon \to 0^+$.

ϵ	$\widehat{\sigma}_c(\epsilon)$	$\sigma_1^* + \frac{76}{9}\epsilon$	$\dfrac{\widehat{\sigma}_c(\epsilon) - \sigma_1^*}{\epsilon}$
0.02000	2.868442	2.835556	10.0888
0.01000	2.758704	2.751111	9.2037
0.00500	2.710719	2.708889	8.8106
0.00250	2.688228	2.687778	8.6243
0.00125	2.677334	2.677222	8.5336

Thus the first correction reduces the discrepancy from approximately 9.20×10^{-2} to 7.59×10^{-3}.

To test the asymptotic coefficient itself, rather than a single finite-ϵ case, we repeat the calculation for a decreasing sequence of ϵ-values. Table 1 reports the numerical crossing, the corrected asymptotic prediction, and the scaled displacement from the leading critical value.

As ϵ decreases, the scaled displacement approaches $76/9$, while the difference

$$\widehat{\sigma}_c(\epsilon) - \left(\sigma_1^* + \frac{76}{9}\epsilon\right)$$

decreases at a rate consistent with the $\mathcal{O}(\epsilon^2)$ remainder in (88). Thus the calculation tests the asymptotic critical-curve expansion quantitatively, rather than merely confirming that a multiplier crossing occurs.

For $\epsilon = 10^{-2}$, the scan $2.55 \le \sigma_1 \le 2.97$ shows that the nontrivial multiplier pair remains nonreal and crosses the unit circle exactly once. Over this scan the minimum absolute imaginary part of the nontrivial pair is 0.3357, and the largest shooting residual is less than 1.3×10^{-13}. Figure 1 shows the corresponding multiplier-modulus crossing.

These computations corroborate both the finite-ϵ multiplier crossing and the predicted displacement of the critical parameter. They are not part of the proof of Theorem 1.3; instead, they provide an independent full-system check of Proposition 3.3. The next subsection complements this multiplier calculation by displaying the post-bifurcation attracting set on the canonical Poincaré section.

4.2. **Numerical signature of the bifurcating invariant torus.** The Floquet calculation locates the loss of transverse stability, but it does not by itself reveal the geometry of the bifurcating invariant set. We therefore examine the full system on the parameter side selected by Theorem 1.3. For

$$\epsilon = 10^{-2}, \qquad \sigma_1 = 2.80,$$

one has

$$\sigma_1 - \widehat{\sigma}_c(10^{-2}) \approx 4.13 \times 10^{-2} > 0.$$

Since $P_{\mathrm{HR}} > 0$ and $L_{\mathrm{HR}} < 0$ in Example 3.16, the theorem predicts a supercritical, normally attracting torus on this side of the critical curve.

At $\sigma_1 = 2.80$, single shooting gives the periodic orbit period

$$T = 6.198892641\ldots.$$

The two nontrivial Floquet multipliers are

$$\mu_\pm = 0.935316438 \pm 0.355721153\, i, \qquad |\mu_\pm| = 1.000676960 > 1,$$

so the periodic orbit has already become transversely unstable.

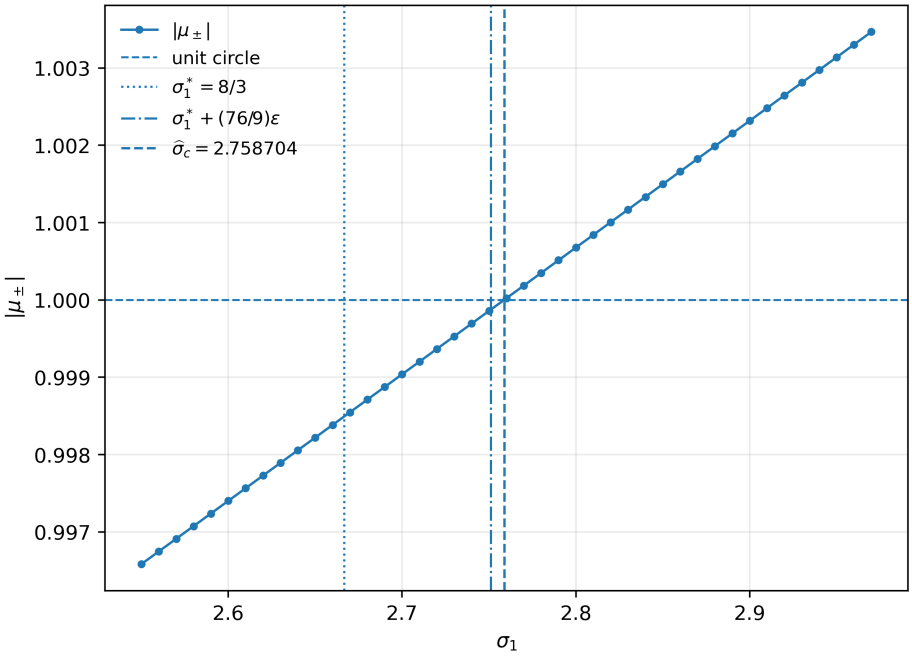


FIGURE 1. Full-system Floquet validation for Example 3.16 at $\epsilon = 10^{-2}$. The modulus of the nontrivial conjugate multiplier pair crosses the unit circle at $\widehat{\sigma}_c = 2.758704047\ldots$. The vertical reference lines show the leading prediction $\sigma_1^* = 8/3$, the first corrected asymptotic prediction $\sigma_1^* + (76/9)\epsilon = 2.751111\ldots$, and the numerical crossing. Throughout the scan the pair remains nonreal; the minimum absolute imaginary part is 0.3357.

To visualize the nearby attracting set in coordinates consistent with the analysis, we use the real canonical variables introduced in Section 2.1. Let

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = P^{-1} \begin{pmatrix} u \\ v \\ w \end{pmatrix},$$

where (u, v, w) are measured from the fixed equilibrium p_ϵ. The theoretical angular variable satisfies $\theta = \operatorname{atan2}(Y, X)$; hence the natural local Poincaré section is

$$\Sigma = \{Y = 0,\ X > 0\}.$$

On Σ we record

$$\rho = \frac{X}{\epsilon}, \qquad \zeta = \frac{Z}{\epsilon}.$$

The fixed point of this return map corresponding to the unstable periodic orbit is located numerically at

$$(\rho_{\mathrm{po}}, \zeta_{\mathrm{po}}) = (2.025314, -1.444959).$$

We initialize a trajectory 10^{-5} away from the periodic orbit and integrate until the section iterates settle onto the attracting set. A point on this settled set is then used to restart an independent high-accuracy DOP853 integration with relative tolerance 10^{-11}, absolute tolerance 10^{-13}, and maximum step 0.2. Over 1.2×10^4 time units, 1938 successive crossings of Σ were recorded. As shown in Figure 2, these iterates closely trace a smooth closed curve surrounding the unstable fixed point rather than collapsing to a finite set. This is the expected Poincaré-section signature of an invariant two-torus and is consistent with quasiperiodic motion on that torus.

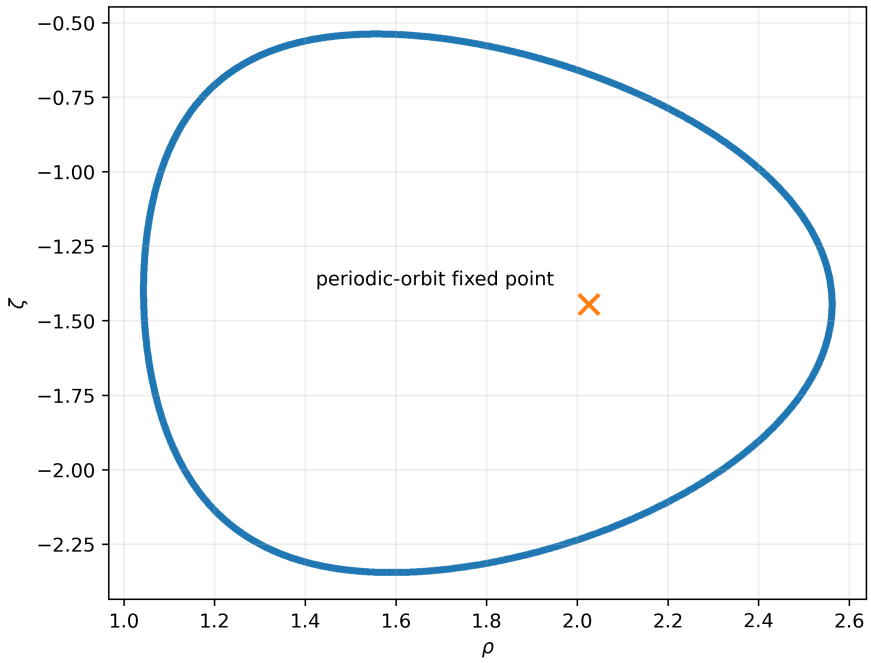


FIGURE 2. Canonical Poincaré-section signature of the bifurcating invariant torus for Example 3.16 with $\epsilon = 10^{-2}$ and $\sigma_1 = 2.80$. The section is $\Sigma = \{Y = 0,\ X > 0\}$, and the plotted coordinates are $(\rho, \zeta) = (X/\epsilon, Z/\epsilon)$. The section iterates form a smooth closed curve around the fixed point (cross) corresponding to the now-unstable periodic orbit. The 1938 plotted crossings were obtained from the high-accuracy DOP853 validation run after the attracting set had been reached.

The numerical rotation number of the closed-curve dynamics, estimated from the lifted polar angle around the periodic-orbit fixed point, is

$$|\varrho_{\mathrm{num}}| \approx 0.0479.$$

The section points remain on the same closed curve under step-size refinement and under the independent adaptive integration described above. Combined with the transverse Floquet instability of the central periodic orbit and the analytically determined sign $L_{\mathrm{HR}} < 0$, the section provides a direct full-system corroboration of the supercritical attracting torus predicted by Theorem 1.3. This calculation validates the local picture; it is not intended as a global continuation of the torus branch.

4.3. **Square-root amplitude scaling near the torus bifurcation.** The local normal form of a supercritical Neimark–Sacker bifurcation gives the amplitude law

$$R = \mathcal{O}\Big(\sqrt{\sigma_1 - \sigma_c(\epsilon)}\Big)$$

on the torus-existence side. The canonical section coordinates (ρ, ζ) are not the final normal-form coordinates, but a smooth nonsingular local coordinate change does not alter the square-root exponent. This allows us to test the scaling directly in the full system.

For each value of $\sigma_1 > \widehat{\sigma}_c(10^{-2})$, we compute the periodic-orbit fixed point $(\rho_{\mathrm{po}}, \zeta_{\mathrm{po}})$ on the canonical section $\Sigma = \{Y = 0,\ X > 0\}$, allow the post-bifurcation trajectory to settle

TABLE 2. Full-system section amplitude of the attracting invariant curve for $\epsilon = 10^{-2}$. Here $\Delta\sigma = \sigma_1 - \widehat{\sigma}_c$, with $\widehat{\sigma}_c = 2.758704047\ldots$.

σ_1	$\Delta\sigma$	A_Σ	A_Σ^2
2.765	0.006296	0.317482	0.100795
2.770	0.011296	0.431217	0.185948
2.775	0.016296	0.523868	0.274438
2.780	0.021296	0.606541	0.367892
2.790	0.031296	0.756850	0.572822
2.800	0.041296	0.897794	0.806033

onto the attracting closed curve, and define the root-mean-square section amplitude

$$A_\Sigma := \left[\frac{1}{N}\sum_{j=1}^{N}\left((\rho_j - \rho_{\mathrm{po}})^2 + (\zeta_j - \zeta_{\mathrm{po}})^2\right)\right]^{1/2}. \tag{89}$$

Table 2 lists the resulting amplitudes.

For the four parameter values closest to the bifurcation, a least-squares fit of A_Σ^2 against σ_1 gives

$$A_\Sigma^2 \approx 17.795603\big(\sigma_1 - 2.759448005\big), \qquad R^2 = 0.999562. \tag{90}$$

The critical parameter inferred solely from the amplitude fit,

$$\sigma_c^{\mathrm{amp}} = 2.759448005,$$

differs from the independently computed Floquet crossing

$$\widehat{\sigma}_c = 2.758704047$$

by only

$$|\sigma_c^{\mathrm{amp}} - \widehat{\sigma}_c| \approx 7.44 \times 10^{-4}.$$

Equivalently, a log–log fit of $A_\Sigma = C(\Delta\sigma)^p$ over the same four closest points gives

$$p = 0.5304, \qquad R^2 = 0.999922,$$

which is close to the Neimark–Sacker normal-form exponent $1/2$.

Figure 3 shows the near-linear dependence of A_Σ^2 on σ_1. The two more distant parameter values are included to display the onset of higher-order deviations away from the local bifurcation regime. Taken together, the Floquet crossing, the attracting invariant closed curve, and the near-square-root growth of its section amplitude provide three complementary numerical signatures of the supercritical Neimark–Sacker bifurcation established analytically in Theorem 1.3.

5. Discussion and conclusions

The analysis carried out in this paper describes a local route from a zero–Hopf equilibrium to an invariant two-torus in the Hindmarsh–Rose system. The starting point is the small periodic orbit created near the zero–Hopf equilibrium, and the main question is what happens to this orbit when the unfolding parameters are varied. Earlier work on the HR system established Hopf and zero–Hopf bifurcations and the corresponding nearby periodic solutions [5]. Torus and torus-canard mechanisms have also been studied in several neuronal models [4], while toroidal behavior in the HR system has been investigated in stochastic settings [21, 22]. Here we follow the periodic orbit created near the zero–Hopf point one step further and show that, under explicit nondegeneracy conditions, it undergoes a Neimark–Sacker bifurcation that creates a local invariant two-torus.

This mechanism is closely related to the Hopf-zero-to-torus scenario obtained for the FitzHugh–Nagumo system by Cândido, Novaes, and Sadri [7]. In both problems, averaging

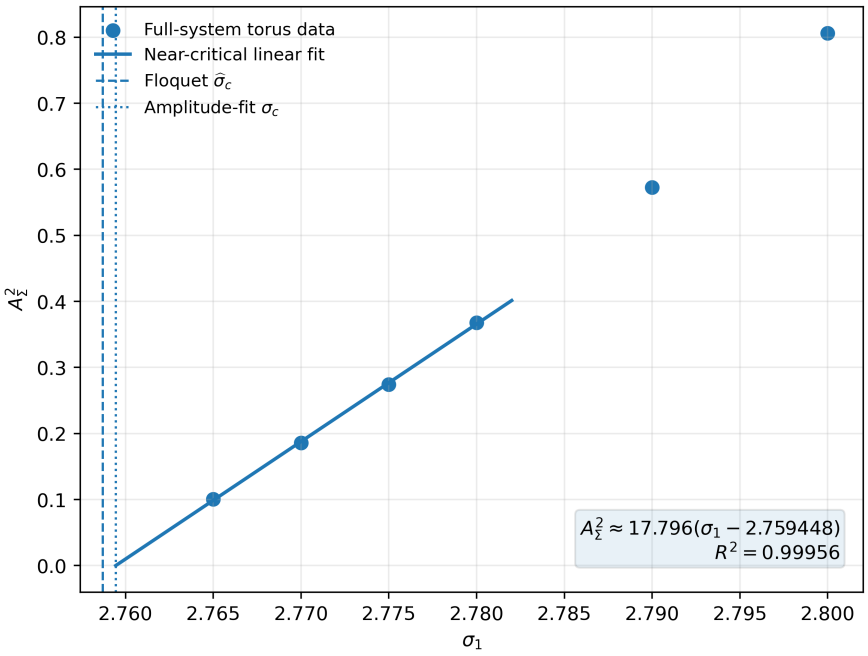


FIGURE 3. Growth of the attracting invariant-curve amplitude on the canonical Poincaré section for $\epsilon = 10^{-2}$. The ordinate is A_Σ^2, defined by (89). The solid line is a linear fit to the four parameter values closest to the bifurcation. The vertical dashed line marks the independently computed Floquet crossing $\widehat{\sigma}_c = 2.758704047\ldots$, while the dotted line marks the critical value extrapolated from the amplitude fit. Their close agreement and the near-linearity of A_Σ^2 are consistent with the expected square-root torus-amplitude law.

first produces a small periodic orbit and the torus is then detected through a Neimark–Sacker bifurcation of the associated Poincaré map. The present HR calculation, however, makes the finite-ϵ location of the bifurcation explicit. The first averaged trace condition identifies only the leading value σ_1^*, whereas the actual transverse multipliers differ from unit modulus at order ϵ^2. Thus the second-order return map is needed not simply to improve an approximation, but to determine the true local critical curve

$$\sigma_c(\epsilon) = \sigma_1^* + \epsilon\sigma_{1,1} + \mathcal{O}(\epsilon^2)$$

on which the nontrivial Floquet multipliers have exactly unit modulus. The same calculation also gives the transverse crossing of the unit circle.

A second feature appears in the normal-form calculation itself. Along the critical curve, the rotation angle of the Poincaré map satisfies

$$\vartheta_\epsilon = \mathcal{O}(\epsilon),$$

so the homological resolvents entering the Neimark–Sacker coefficient become singular as $\epsilon \to 0$. This dependence cannot be discarded. Keeping it throughout the calculation reveals the cancellation of the nominal order-ϵ contribution to the first Lyapunov coefficient and leads to

$$l_1^\epsilon = \epsilon^2 L_{\mathrm{HR}} + \mathcal{O}(\epsilon^3).$$

The factor L_{HR} therefore gives the first nonvanishing criticality coefficient for the unfolding considered here. Because it can be written explicitly in terms of the HR parameters, its sign determines whether the bifurcation is supercritical or subcritical and, accordingly, whether the bifurcating torus is normally attracting or normally repelling.

The numerical calculations were designed to check these analytical predictions directly in the full three-dimensional system. They provide three complementary tests. First, the numerically computed Floquet crossing approaches the asymptotic critical curve with the predicted finite-ϵ displacement. Second, on the parameter side selected by Theorem 1.3, the periodic-orbit fixed point on the canonical Poincaré section becomes unstable and is surrounded by an attracting invariant closed curve. Third, the amplitude of this closed curve grows approximately as the square root of the distance from the critical parameter, in agreement with the local Neimark–Sacker normal form. These computations therefore test the location, geometry, and criticality of the bifurcation separately. They are intended as independent full-system checks of the analysis; the existence of the torus itself follows from the analytical Neimark–Sacker argument.

There are also clear limitations to the result. The most important one concerns the parameter regime. Under the usual positivity assumptions, the zero–Hopf relations used here are incompatible with the standard slow-adaptation HR regime. The theorem should therefore be interpreted as a rigorous result for the generalized polynomial HR family selected by the zero–Hopf conditions, rather than as a direct physiological statement about the conventional neuronal parameter set. The analysis is also local: it describes the torus close to its birth, but does not follow it to large amplitude, determine resonance tongues, or describe torus breakdown and the subsequent transition to more complicated dynamics. In addition, the numerical validation is carried out for one explicit algebraic parameter regime. Finally, when

$$P_{\mathrm{HR}} = 0,$$

the order-ϵ^2 Lyapunov coefficient vanishes, and a higher-order normal-form calculation would be required.

These limitations point naturally to several directions for further work. One question is whether a modified HR-type family can realize the same zero–Hopf-to-torus mechanism while remaining inside a conventional slow-adaptation parameter regime. A second is to continue the local torus numerically and investigate how it is connected to resonance tongues, torus-canard behavior, and torus breakdown. The degenerate case $P_{\mathrm{HR}} = 0$ is another natural problem, since it represents a higher-codimension situation in which the leading criticality coefficient no longer determines the local dynamics.

In summary, the local scenario established here is

$$\begin{aligned} \text{zero–Hopf equilibrium} &\longrightarrow \text{small periodic orbit} \\ &\longrightarrow \text{exact finite-}\epsilon\text{ unit-modulus crossing} \\ &\longrightarrow \text{nondegenerate invariant two-torus.} \end{aligned}$$

The main contribution is therefore not the observation of quasiperiodic behavior in the HR model by itself. Rather, it is the higher-order analytical construction that identifies where the torus bifurcation occurs, proves the required nondegeneracy and crossing conditions, determines the side and normal stability of the bifurcating torus, and connects these predictions with quantitative computations in the full system.

Appendix A. Exact symbolic verification of the Neimark–Sacker coefficient

This appendix completes the proof of Lemma 3.7 by deriving the explicit first Neimark–Sacker Lyapunov coefficient from the second-order Poincaré-map expansion obtained in Section 3. The calculation is organized in four steps:

(1) determination of the first correction to the critical fixed point and to the exact unit-modulus curve;
(2) construction of the quadratic and cubic Taylor tensors of the centered Poincaré map;

(3) expansion of the homological resolvents appearing in the Neimark–Sacker normal-form formula;
(4) symbolic extraction and simplification of the first Lyapunov coefficient.

Unless stated otherwise, all quantities in this appendix are evaluated at the critical point (q_*, σ_1^*).

For convenience introduce the notation

$$A := A_* = D_q g_1(q_*; \sigma_1^*), \qquad H_1 := D_q^2 g_1, \qquad H_2 := D_q^2 g_2, \qquad T_2 := D_q^3 g_2,$$

together with

$$g_{1,\sigma} := \partial_{\sigma_1} g_1, \qquad A_\sigma := \partial_{\sigma_1} D_q g_1.$$

The critical averaged identities established in Section 3 are

$$\zeta_* = -\frac{A_X}{2B_X}, \qquad \rho_*^2 = \frac{A_X^2}{2B_X^2}, \qquad A_Z = \frac{A_X B_Z}{B_X}, \qquad \nu_0^2 = -\frac{B_X B_Z \rho_*^2}{\omega^2}. \tag{91}$$

Consequently,

$$A = \frac{\rho_*}{\omega}\begin{pmatrix} 0 & B_X \\ B_Z & 0 \end{pmatrix}, \qquad A^2 = -\nu_0^2 I.$$

A.1. **First correction of the critical fixed point.** The branch of critical fixed points constructed in Lemma 3.2 has the expansion

$$\widehat{q}_\epsilon = q_* + \epsilon u + \mathcal{O}(\epsilon^2), \qquad u = (u_\rho, u_\zeta)^{\mathsf{T}}.$$

Similarly, the exact unit-modulus curve obtained in Proposition 3.3 satisfies

$$\sigma_c(\epsilon) = \sigma_1^* + \epsilon\sigma_{1,1} + \mathcal{O}(\epsilon^2).$$

The unknowns $u_\rho, u_\zeta, \sigma_{1,1}$ are determined simultaneously by the order-ϵ fixed-point equation together with the order-ϵ^2 unit-determinant condition:

$$Au + \sigma_{1,1} g_{1,\sigma} + g_2 = 0, \qquad \operatorname{tr} A_2 + 2\pi \det A = 0, \tag{92}$$

where

$$A_2 = H_1[u, \cdot] + \sigma_{1,1} A_\sigma + D_q g_2. \tag{93}$$

The coefficient matrix of (92) is nonsingular by Lemma 3.2 together with the transversality condition $\tau'(\sigma_1^*) \neq 0$. Hence the system uniquely determines $(u, \sigma_{1,1})$. The linear part of the centered Poincaré map therefore has the expansion

$$L_\epsilon = I + 2\pi\epsilon A + 2\pi\epsilon^2 A_2 + \mathcal{O}(\epsilon^3). \tag{94}$$

A.2. **Quadratic and cubic Taylor tensors.** Since the first averaged function g_1 is quadratic, $D_q^3 g_1 = 0$. Moreover, $\partial_{\sigma_1} D_q^2 g_1 = 0$, because the coefficients B_X and B_Z are independent of the bifurcation parameter. Differentiating the second-order expansion of the Poincaré map therefore gives

$$B_\epsilon = 2\pi\epsilon H_1 + 2\pi\epsilon^2 H_2 + \mathcal{O}(\epsilon^3), \qquad C_\epsilon = 2\pi\epsilon^2 T_2 + \mathcal{O}(\epsilon^3). \tag{95}$$

The displacement u does not appear explicitly in these tensor expansions. Its effect is contained in the linear correction matrix A_2 and therefore enters the normal-form calculation through the linear conjugacy.

To transform the linear part into rotation form, write

$$\vartheta_\epsilon = 2\pi\nu_0\epsilon + \theta_2\epsilon^2 + \mathcal{O}(\epsilon^3),$$

where the second-order coefficient θ_2 is left undetermined. Introduce

$$b_{\mathrm{rot},0} = 2\pi A_{12}, \qquad b_{\mathrm{rot},1} = 2\pi (A_2)_{12},$$

and define

$$u_S = \frac{-2\pi^2\nu_0^2 - 2\pi(A_2)_{11}}{b_{\mathrm{rot},0}}, \quad v_0 = -\frac{2\pi\nu_0}{b_{\mathrm{rot},0}},$$
$$v_1 = -\frac{\theta_2}{b_{\mathrm{rot},0}} + \frac{2\pi\nu_0 b_{\mathrm{rot},1}}{b_{\mathrm{rot},0}^2}, \quad S_0 = \begin{pmatrix} 1 & 0 \\ 0 & v_0 \end{pmatrix}, \quad S_1 = \begin{pmatrix} 0 & 0 \\ u_S & v_1 \end{pmatrix}.$$

Let

$$T_0 = S_0^{-1}, \qquad T_1 = \begin{pmatrix} 0 & 0 \\ -u_S/v_0 & -v_1/v_0^2 \end{pmatrix}.$$

Expansion of the exact conjugacy matrix in (77) gives

$$S_\epsilon = S_0 + \epsilon S_1 + \mathcal{O}(\epsilon^2), \qquad S_\epsilon^{-1} = T_0 + \epsilon T_1 + \mathcal{O}(\epsilon^2).$$

For vectors $x, y, z \in \mathbb{C}^2$, define

$$U_1(x,y) = 2\pi H_1[x,y], \qquad U_2(x,y) = 2\pi H_2[x,y], \qquad V_2(x,y,z) = 2\pi T_2[x,y,z].$$

Substitution into the transformation formulas for multilinear tensors yields

$$\widetilde{B}_\epsilon = \epsilon B_0 + \epsilon^2 B_1 + \mathcal{O}(\epsilon^3), \qquad \widetilde{C}_\epsilon = \epsilon^2 C_0 + \mathcal{O}(\epsilon^3), \tag{96}$$

where

$$B_0(x,y) = T_0 U_1(S_0 x, S_0 y),$$
$$B_1(x,y) = T_1 U_1(S_0 x, S_0 y) + T_0 U_1(S_1 x, S_0 y) + T_0 U_1(S_0 x, S_1 y) + T_0 U_2(S_0 x, S_0 y),$$
$$C_0(x,y,z) = T_0 V_2(S_0 x, S_0 y, S_0 z).$$

Equations (92), (94), and (96) contain every quantity that contributes to the first Lyapunov coefficient through order ϵ^2.

A.3. **Homological resolvents.** Let

$$q = \begin{pmatrix} 1 \\ -i \end{pmatrix}, \qquad p_{\mathrm{ad}} = \frac{1}{2}\begin{pmatrix} 1 \\ -i \end{pmatrix}.$$

The linear part of the centered map is the rotation R_{ϑ_ϵ}, whose angle satisfies

$$\vartheta_\epsilon = 2\pi\nu_0\epsilon + \theta_2\epsilon^2 + \mathcal{O}(\epsilon^3).$$

Accordingly,

$$R_{\vartheta_\epsilon} = I + \epsilon J_1 + \epsilon^2 R_2 + \mathcal{O}(\epsilon^3),$$

where

$$J_1 = \begin{pmatrix} 0 & -2\pi\nu_0 \\ 2\pi\nu_0 & 0 \end{pmatrix}, \qquad R_2 = \begin{pmatrix} -2\pi^2\nu_0^2 & -\theta_2 \\ \theta_2 & -2\pi^2\nu_0^2 \end{pmatrix}.$$

The homological equations defining the normal-form transformation are

$$(I - R_{\vartheta_\epsilon})h_{11} = \widetilde{B}_\epsilon(q,\overline{q}), \qquad (\lambda_\epsilon^2 I - R_{\vartheta_\epsilon})h_{20} = \widetilde{B}_\epsilon(q,q).$$

Although the corresponding linear operators become singular as $\epsilon \to 0$, the singularity is exactly compensated by the fact that $\widetilde{B}_\epsilon = \mathcal{O}(\epsilon)$. Consequently both homological vectors admit finite asymptotic expansions

$$h_{11} = h_{11}^{(0)} + \epsilon h_{11}^{(1)} + \mathcal{O}(\epsilon^2), \qquad h_{20} = h_{20}^{(0)} + \epsilon h_{20}^{(1)} + \mathcal{O}(\epsilon^2).$$

Substituting the tensor expansions above gives

$$h_{11}^{(0)} = (-J_1)^{-1} B_0(q,\overline{q}),$$
$$h_{11}^{(1)} = (-J_1)^{-1}\big(B_1(q,\overline{q}) + R_2 h_{11}^{(0)}\big),$$
$$h_{20}^{(0)} = N_1^{-1} B_0(q,q),$$
$$h_{20}^{(1)} = N_1^{-1}\big(B_1(q,q) - N_2 h_{20}^{(0)}\big),$$

where

$$N_1 = 4\pi i\nu_0 I - J_1, \qquad N_2 = (2i\theta_2 - 8\pi^2\nu_0^2)I - R_2.$$

Since $\nu_0 > 0$, both $-J_1$ and N_1 are nonsingular.

A.4. **Symbolic extraction of the first Lyapunov coefficient.** Define

$$\begin{aligned}\mathfrak{N}_1 &= 2B_0(q, h_{11}^{(0)}) + B_0(\overline{q}, h_{20}^{(0)}),\\ \mathfrak{N}_2 &= C_0(q, q, \overline{q}) + 2B_0(q, h_{11}^{(1)}) + 2B_1(q, h_{11}^{(0)})\\ &\quad + B_0(\overline{q}, h_{20}^{(1)}) + B_1(\overline{q}, h_{20}^{(0)}).\end{aligned}$$

Substitution into the exact Neimark–Sacker coefficient formula (80) gives

$$l_1^{\epsilon} = \epsilon\ell_{1,1} + \epsilon^2\ell_{1,2} + \mathcal{O}(\epsilon^3),$$

where

$$\ell_{1,1} = \frac{1}{2}\,\mathrm{Re}\langle p_{\mathrm{ad}}, \mathfrak{N}_1\rangle,$$

and

$$\ell_{1,2} = \frac{1}{2}\,\mathrm{Re}\left(\langle p_{\mathrm{ad}}, \mathfrak{N}_2\rangle - 2\pi i\nu_0\langle p_{\mathrm{ad}}, \mathfrak{N}_1\rangle\right).$$

The order-ϵ cancellation can be shown explicitly, rather than left as an opaque computer-algebra simplification. Set

$$\chi := v_0 = -\frac{2\pi\nu_0}{b_{\mathrm{rot},0}}.$$

Since $b_{\mathrm{rot},0} = 2\pi\rho_* B_X/\omega$, the critical identities give

$$\chi^2 = -\frac{B_Z}{B_X}, \qquad \nu_0 = -\frac{\rho_* B_X}{\omega}\chi.$$

Using $S_0 = \mathrm{diag}(1, \chi)$ in the definition of B_0 gives

$$B_0(q, \overline{q}) = \begin{pmatrix} 0 \\ -\dfrac{2\pi B_X\chi}{\omega}(2\chi^2 + 1) \end{pmatrix},$$

and

$$B_0(q, q) = \begin{pmatrix} -\dfrac{4\pi i B_X\chi}{\omega} \\ \dfrac{2\pi B_X\chi}{\omega}(2\chi^2 - 1) \end{pmatrix}.$$

The leading homological vectors are therefore

$$h_{11}^{(0)} = \begin{pmatrix} -\dfrac{2\chi^2 + 1}{\rho_*} \\ 0 \end{pmatrix}, \qquad h_{20}^{(0)} = \begin{pmatrix} \dfrac{5 - 2\chi^2}{3\rho_*} \\ \dfrac{4i(\chi^2 - 1)}{3\rho_*} \end{pmatrix}.$$

A direct substitution then yields

$$\langle p_{\mathrm{ad}}, \mathfrak{N}_1\rangle = \frac{4\pi i B_X\chi}{3\rho_*\omega}\left(2\chi^4 + 5\chi^2 + 2\right). \tag{97}$$

The right-hand side is purely imaginary because $B_X, \chi, \rho_*, \omega \in \mathbb{R}$. Hence

$$\boxed{\ell_{1,1} = 0}.$$

Thus the first nonzero contribution can occur only at order ϵ^2.

For the order-ϵ^2 coefficient, substitute the polynomial expressions (59)–(60) into $H_2 = D_q^2 g_2$ and $T_2 = D_q^3 g_2$, solve (92) for $(u_\rho, u_\zeta, \sigma_{1,1})$, and insert the resulting

$A_2, B_1, C_0, h_{11}^{(1)}, h_{20}^{(1)}$ into $\mathfrak{N}_2$. Simplification with the critical identities (91) eliminates the auxiliary quantities

$$C_X, C_Y, C_Z, \quad D_X, D_Y, D_Z, \quad E_Y, E_Z, \quad \sigma_{1,1}, \quad \theta_2,$$

and leaves a coefficient depending only on $B_X, B_Y, B_Z, E_X, \omega$:

$$\ell_{1,1} = 0, \qquad \ell_{1,2} = -\frac{\pi B_Z(B_X + B_Z)}{2B_X^2\omega^2}\big(2B_XB_Y - 7B_YB_Z - 3E_X\omega\big). \tag{98}$$

Consequently,

$$l_1^\epsilon = \epsilon^2 L_{\mathrm{HR}} + \mathcal{O}(\epsilon^3),$$

where L_{HR} is exactly (82).

Two features of this reduction are important. First, $\ell_{1,1} = 0$ is a structural cancellation of the leading quadratic normal-form contribution, as (97) shows; it is not a numerical coincidence. Second, $\ell_{1,2}$ is independent of β_2 and σ_2 because the second-order unfolding coefficients contained in C_X, C_Y, C_Z cancel from the final expression.

A.5. **Reduction to the Hindmarsh–Rose parameters.** Substituting

$$B_X = \frac{D_0}{\omega^2}, \qquad B_Y = \frac{d - (1+r)k_0}{\omega}, \qquad B_Z = \frac{r(k_0 - d)}{\omega^2}, \qquad E_X = -\frac{a(\omega^2 - r)}{\omega^2}$$

into (98), and using

$$D_0 + r(k_0 - d) = k_0\omega^2,$$

gives

$$L_{\mathrm{HR}} = -\frac{\pi k_0 r(k_0 - d)}{2\omega^3 D_0^2}\left[3a\omega^2(\omega^2 - r) + \big(2k_0\omega^2 + 9r(d - k_0)\big)\big(d - k_0(1+r)\big)\right].$$

This is exactly the factorized formula (83).

Remark A.1 (Reproducibility). The accompanying symbolic-verification script uses exact SymPy arithmetic for three complementary checks. First, before any specialization to an HR parameter set, it verifies the structural cancellation $\ell_{1,1} = 0$ in (97). Second, it reconstructs the complete order-ϵ^2 normal-form calculation for Example 3.16, including $\sigma_{1,1} = 76/9$, and obtains

$$\ell_{1,2} = -\frac{2016\pi}{121};$$

Third, it verifies identically the algebraic reduction from (82) to (83). The full-system shooting and variational calculations in Section 4 are independent of these symbolic checks.

These scripts serve as reproducibility and error-detection tools. The proof remains the analytical expansion given above; in particular, neither the uniform remainder estimates nor the application of the Neimark–Sacker theorem is delegated to computer algebra.

References

1. Roberto Barrio, Santiago Ibáñez, and Lucía Pérez, *Hindmarsh–Rose model: Close and far to the singular limit*, Physics Letters A **381** (2017), no. 6, 597–603, doi:10.1016/j.physleta.2016.12.027.
2. ———, *Homoclinic organization in the Hindmarsh–Rose model: A three-parameter study*, Chaos **30** (2020), no. 5, 053132, doi:10.1063/1.5138919.
3. ———, *Exploring the geometry of the bifurcation sets in parameter space*, Scientific Reports **14** (2024), 10900, doi:10.1038/s41598-024-61574-6.
4. John Burke, Mathieu Desroches, Anna M. Barry, Tasso J. Kaper, and Mark A. Kramer, *A showcase of torus canards in neuronal bursters*, The Journal of Mathematical Neuroscience **2** (2012), 3, doi:10.1186/2190-8567-2-3.
5. Claudio Buzzi, Jaume Llibre, and João Medrado, *Hopf and zero–hopf bifurcations in the Hindmarsh–Rose system*, Nonlinear Dynamics **83** (2016), no. 3, 1549–1556, doi:10.1007/s11071-015-2429-y.
6. Murilo R Candido and Douglas D Novaes, *On the torus bifurcation in averaging theory*, Journal of Differential Equations **268** (2020), no. 8, 4555–4576.

7. Murilo R Cândido, Douglas D Novaes, and Nasrin Sadri, *Invariant tori and periodic orbits in the fitzhugh-nagumo system*, Nonlinear Analysis: Real World Applications **93** (2027), 104676.
8. Murilo R Cândido, Douglas D Novaes, and Claudia Valls, *Periodic solutions and invariant torus in the rössler system*, Nonlinearity **33** (2020), no. 9, 4512–4539.
9. Jinyi Chen and Lijuan Ning, *Bifurcations and resonances in a birhythmic Hindmarsh–Rose neuron model with different spikes*, Physical Review E **111** (2025), no. 6, 064211, doi:10.1103/zjdf-h43s.
10. Jean-Marc Ginoux, Jaume Llibre, and Kiyoyuki Tchizawa, *Canards existence in the Hindmarsh–Rose model*, Mathematical Modelling of Natural Phenomena **14** (2019), no. 4, 409, doi:10.1051/mmnp/2019012.
11. J. L. Hindmarsh and R. M. Rose, *A model of neuronal bursting using three coupled first order differential equations*, Proceedings of the Royal Society of London. Series B, Biological Sciences **221** (1984), no. 1222, 87–102, doi:10.1098/rspb.1984.0024.
12. Giacomo Innocenti, Alice Morelli, Roberto Genesio, and Alessandro Torcini, *Dynamical phases of the Hindmarsh–Rose neuronal model: Studies of the transition from bursting to spiking chaos*, Chaos **17** (2007), no. 4, 043128, doi:10.1063/1.2818153.
13. Eugene M. Izhikevich, *Dynamical systems in neuroscience: The geometry of excitability and bursting*, MIT Press, Cambridge, MA, 2007, doi:10.7551/mitpress/2526.001.0001.
14. Huiwen Ju, Alexander B. Neiman, and Andrey L. Shilnikov, *Bottom-up approach to torus bifurcation in neuron models*, Chaos **28** (2018), no. 10, 106317, doi:10.1063/1.5042078.
15. Yuri A. Kuznetsov, *Elements of applied bifurcation theory*, 3 ed., Springer, New York, 2004, doi:10.1007/978-1-4757-3978-7.
16. Xinyi Lin, Yaru Liu, and Shenquan Liu, *Torus bifurcation and homoclinic chaos underlie potassium-driven epileptiform dynamics in a multi-timescale neural mass model*, Chaos, Solitons & Fractals **208** (2026), 118204, doi:10.1016/j.chaos.2026.118204.
17. Daniele Linaro, Alan Champneys, Mathieu Desroches, and Marco Storace, *Codimension-two homoclinic bifurcations underlying spike adding in the Hindmarsh–Rose burster*, SIAM Journal on Applied Dynamical Systems **11** (2012), no. 3, 939–962, doi:10.1137/110848931.
18. Jaume Llibre and Jaime Rezende De Moraes, *Analytic study of two limit cycles bifurcating from a zero–hopf equilibrium*, Boletin de la Sociedad Matematica Mexicana **31** (2025), no. 2, 43, doi:10.1007/s40590-025-00726-8.
19. Douglas D Novaes and Pedro CCR Pereira, *Normal hyperbolicity in secondary hopf bifurcations*, Matemática Contemporânea (2026), 1–9.
20. Ignacio Ortega-Piwonka, Javier Used, Jesús M. Seoane, and Miguel A. F. Sanjuán, *Stochastic birhythmicity outside the coexistence region in the Hindmarsh–Rose model*, Physical Review E **111** (2025), no. 4, 044217, doi:10.1103/PhysRevE.111.044217.
21. Lev Ryashko and Evdokia Slepukhina, *Noise-induced torus bursting in the stochastic Hindmarsh–Rose neuron model*, Physical Review E **96** (2017), no. 3, 032212, doi:10.1103/PhysRevE.96.032212.
22. Lev Ryashko and Evdokiia Slepukhina, *Noise-induced toroidal excitability in neuron model*, Communications in Nonlinear Science and Numerical Simulation **82** (2020), 105071, doi:10.1016/j.cnsns.2019.105071.
23. Jan A. Sanders, Ferdinand Verhulst, and James Murdock, *Averaging methods in nonlinear dynamical systems*, 2 ed., Springer, New York, 2007, doi:10.1007/978-0-387-48918-6.
24. Xiaohua Shi, Yi Yang, Xiangguang Dai, Changcheng Xiang, and Yuming Feng, *Andronov–hopf and bogdanov–takens bifurcations in a filippov Hindmarsh–Rose system with switching policy for the slow variable*, Physica D: Nonlinear Phenomena **466** (2024), 134217, doi:10.1016/j.physd.2024.134217.
25. Marco Storace, Daniele Linaro, and Enno de Lange, *The Hindmarsh–Rose neuron model: Bifurcation analysis and piecewise-linear approximations*, Chaos **18** (2008), no. 3, 033128, doi:10.1063/1.2975967.
26. Yi Yang, Jiajia Jiang, Xiaohua Shi, and Wenjie Qin, *Bifurcations and the switching of spiking and bursting in the extended Hindmarsh–Rose neuronal system*, Advances in Continuous and Discrete Models **2026** (2026), 63, doi:10.1186/s13662-026-04101-3.